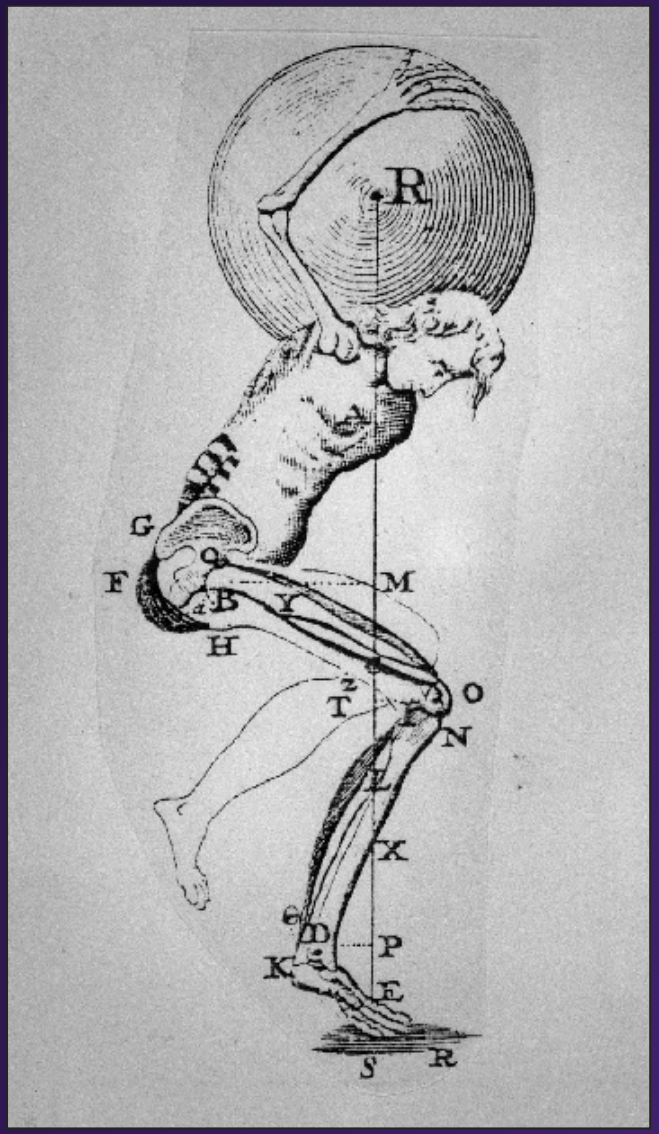


Musculoskeletal Biomechanics

BIOEN 520 | ME 527

Session 14A1

Biomechanics of the Hip

Session 13 Review…

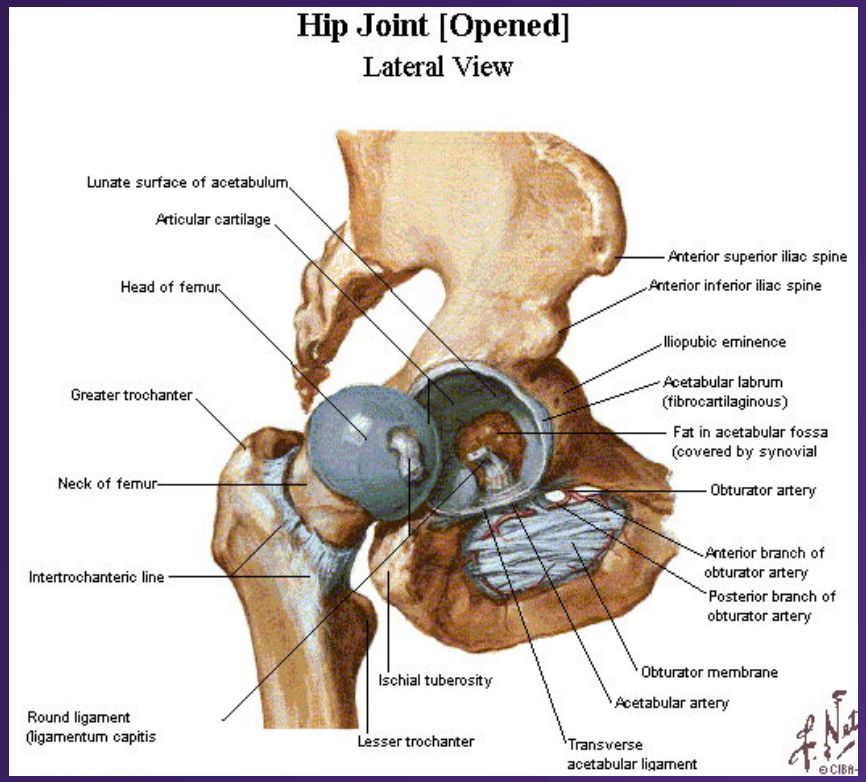
- Gait lab tour and demo
- Biomechanical testing lab tour and demo
- Lab 2 assignment

Session 14A1 Overview…

- Hip
- Anatomy
- General function
- Pathology

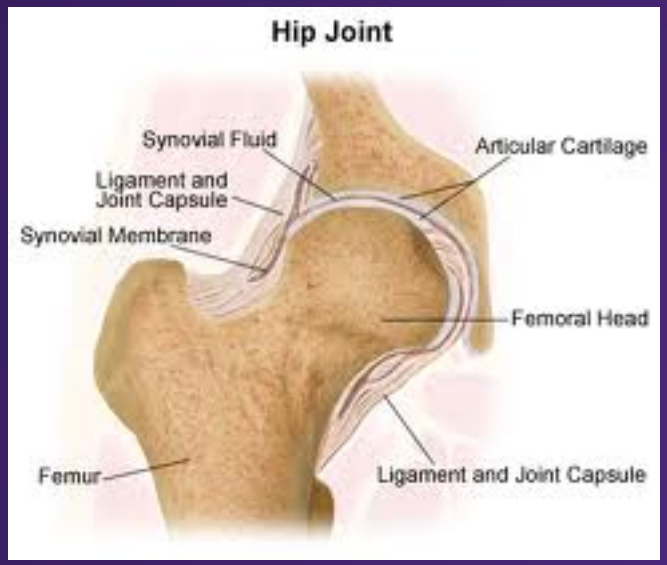
Hip: Anatomy

- type of joint:
  - synovial (diarthrodial)
  - ball and socket



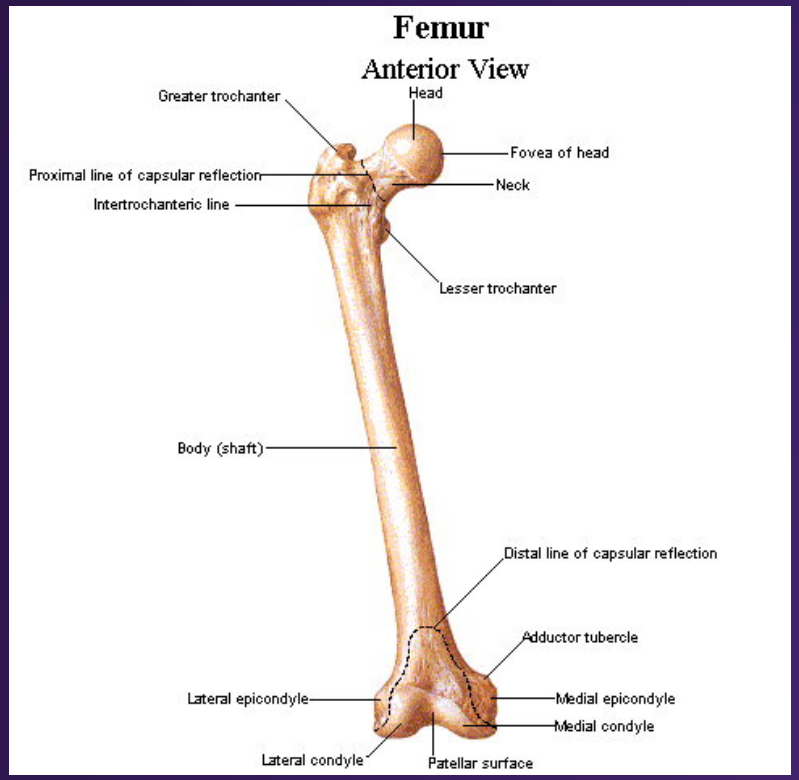
Hip: Anatomy

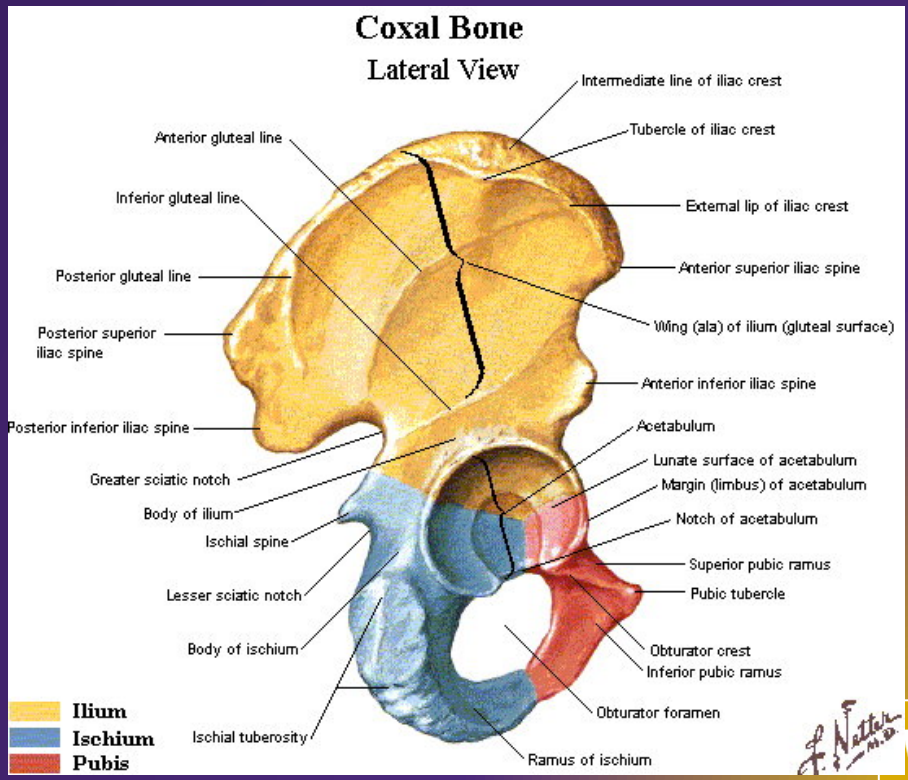
- synovial joint:
  - ends of bones lined with articular cartilage
  - encapsulated with a fibrous capsule
  - capsule lined with synovium
  - stabilized with ligaments and tendons



Hip: Anatomy

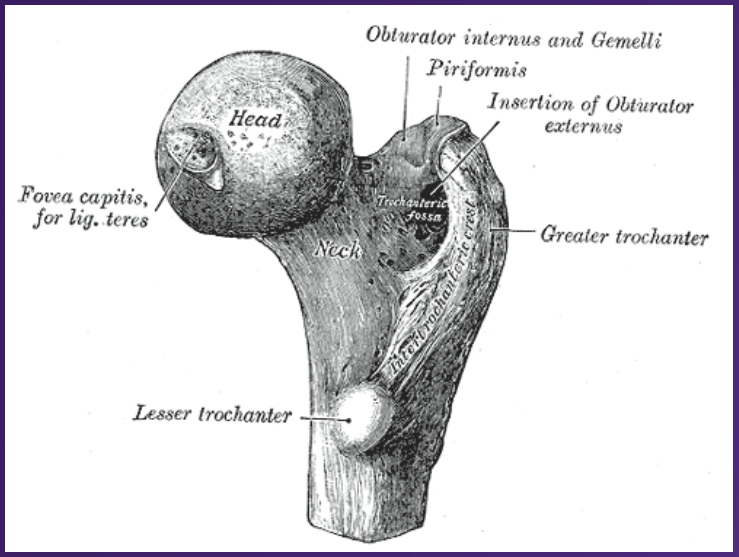
- bones:
  - femur
  - pelvis (ilium, ischium, pubis)





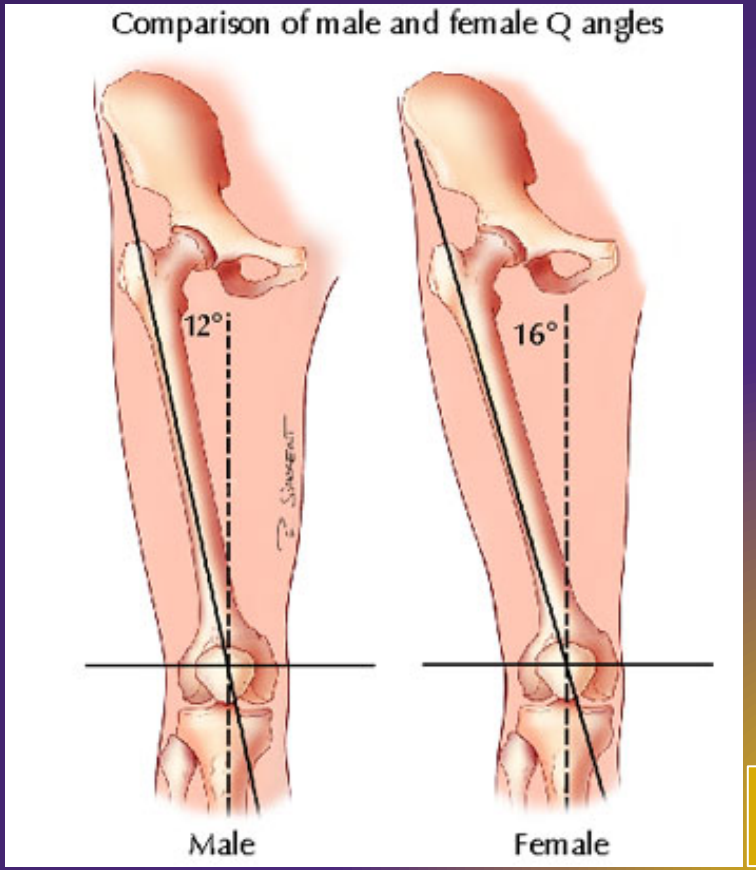
Hip: Anatomy

- femur:
  - femoral head is spherical
    - superior and medial (125˚)
    - anteversion (transverse, 15˚ to 20˚)
  - femoral neck is obliquely orientated



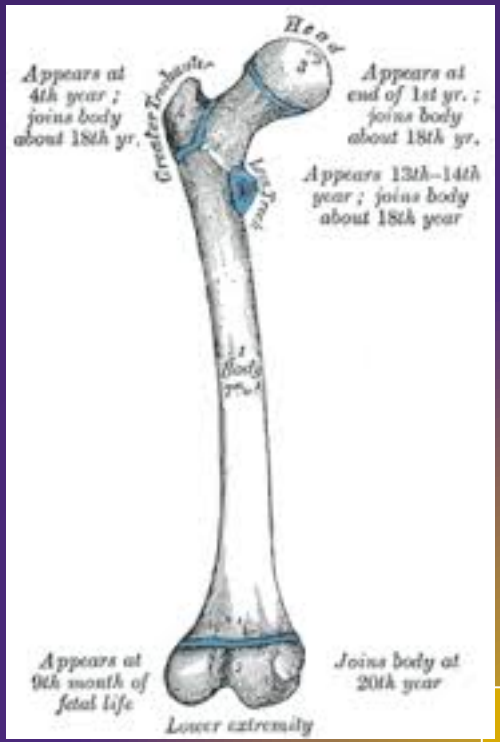
Hip: Anatomy

- femur:
  - Q-angle
    - ASIS to patella, subtract from 180˚
  - male (12˚)
  - female (16˚)



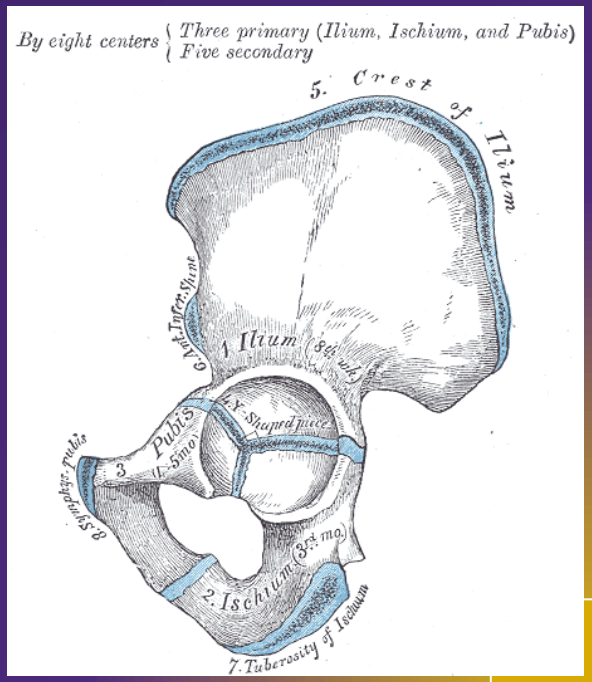
Hip: Anatomy

- femur:
  - longest, heaviest, strongest
  - 26% of person's height
  - useful for estimating heights
  - greater and lesser trocanter



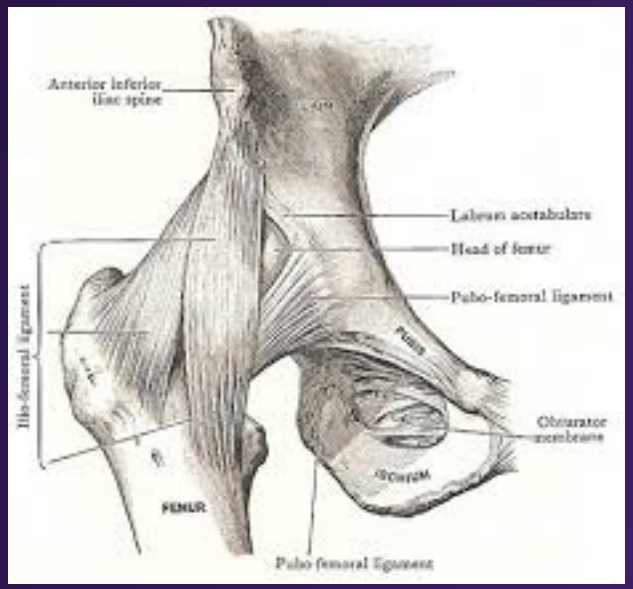
Hip: Anatomy

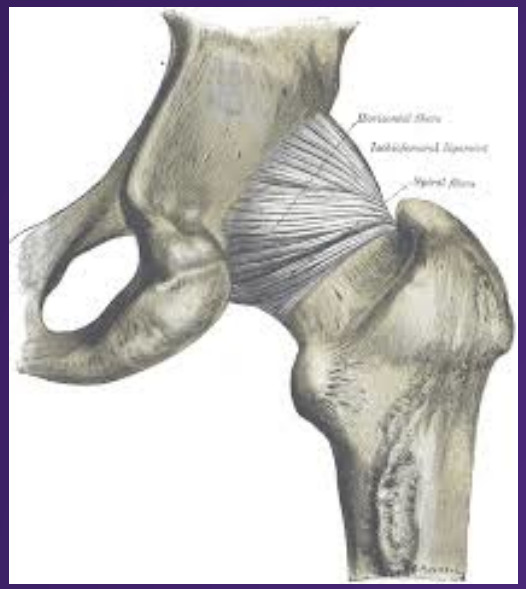
- pelvis:
  - largest bone in the body (6 bones)
  - acetabulum is a concave structure
    - anterior, lateral and inferior
  - ossification center in middle

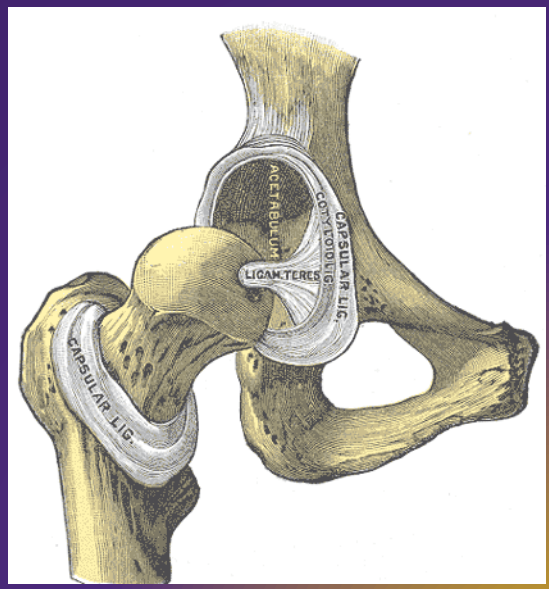


Hip: Anatomy

- capsule: two directions of fibers
  - distal and lateral from pelvis to femur
  - annular fibers (zona orbicularis)

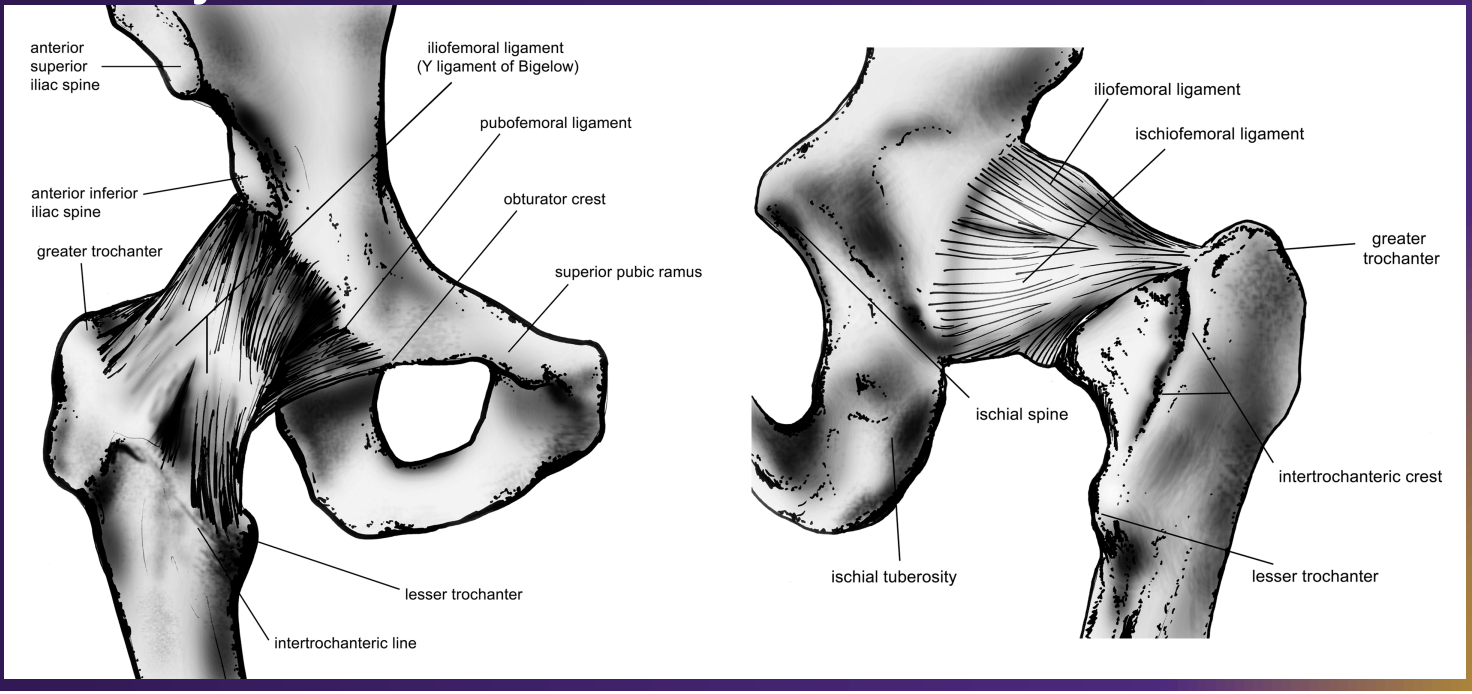






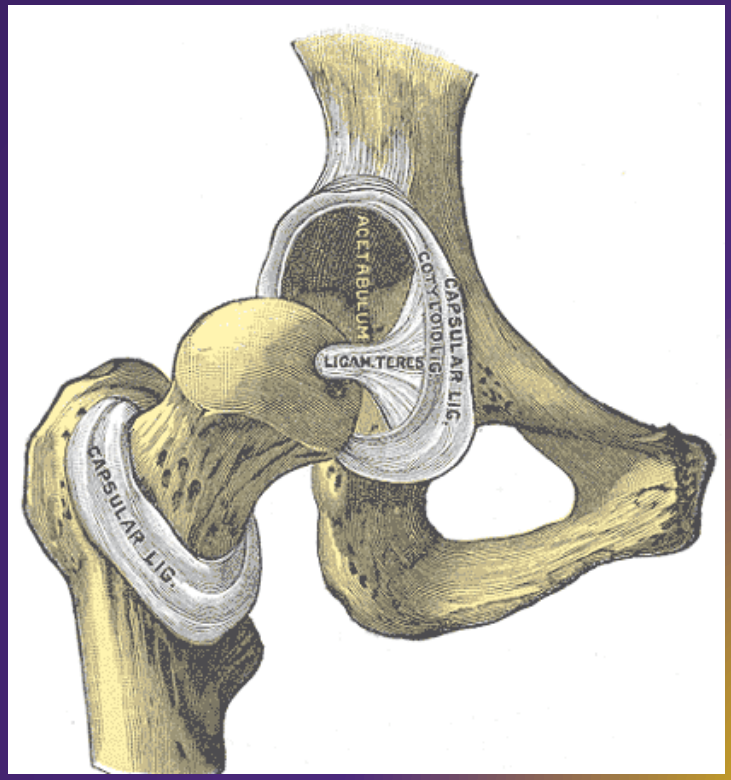
Hip: Anatomy

- ligaments:
  - iliofemoral and pubofemoral
    - iliofemoral: strongest ligament = 350 tensile strength
  - ischiofemoral
  - limit joint motion



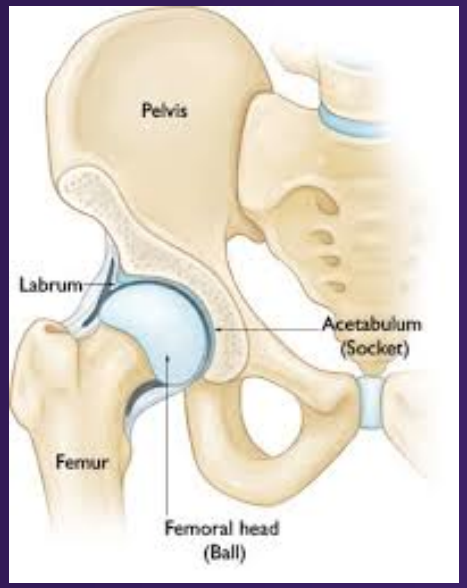
Hip: Anatomy

- ligaments:
  - ligamentum teres
    - protects artery
  - zona orbicularis
    - button hole – little stability?
    - stability in distraction?
    - not well understood



Hip: Anatomy

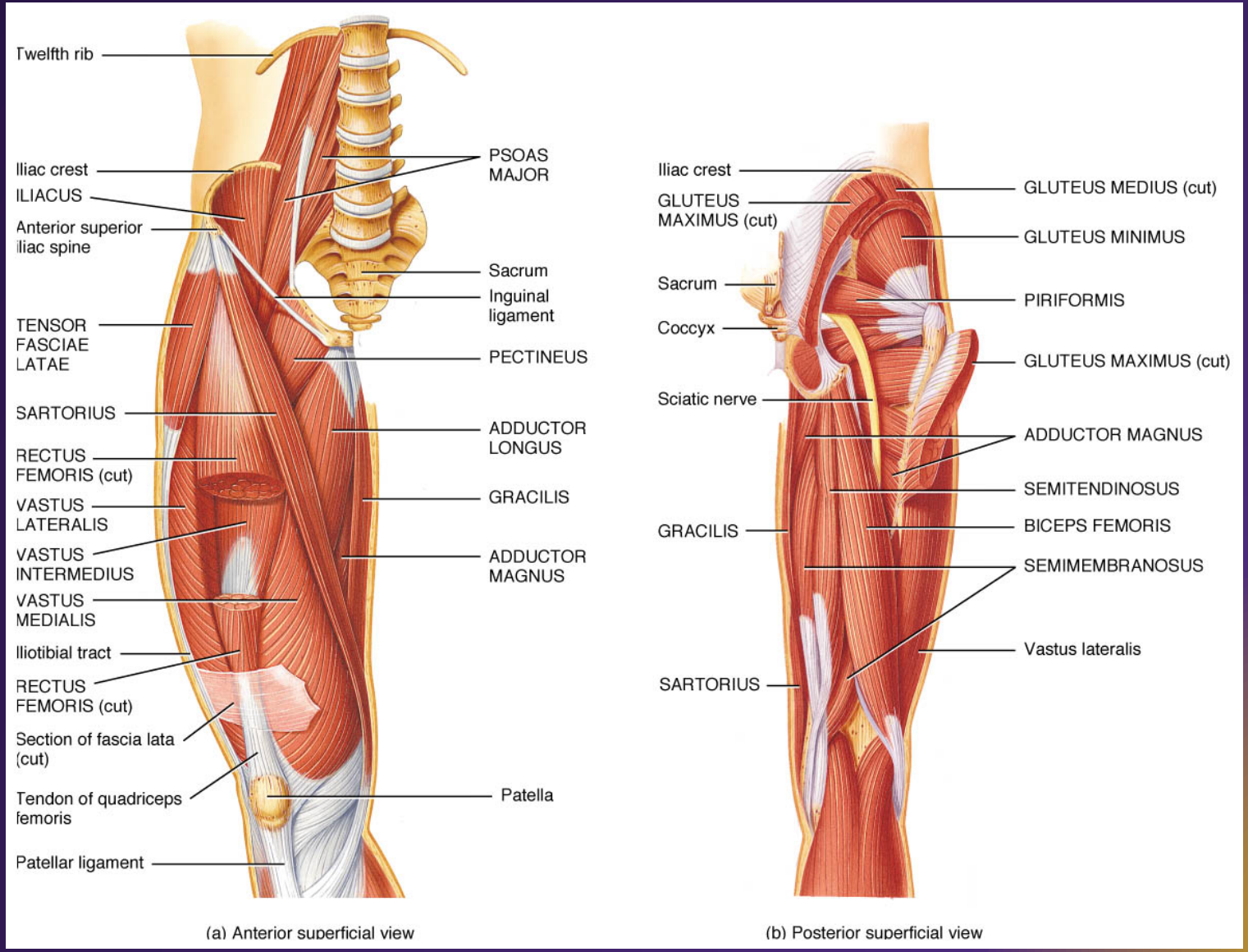
- cartilage:
  - thickest on anterosuperior acetabulum and anterolateral femoral head
  - labrum – fibrocartilage ring, stabilizing all but inferior rim



Hip: Anatomy

- muscles:
  - 22 acting around hip joint
  - sagittal plane - flexion and extension
  - frontal plane – abduction and adduction
  - transverse plane – internal and external rotation

Hip: Anatomy


Twelfth rib
Iliac crest
ILIACUS
Anterior superior iliac spine
TENSOR FASCIAE LATAE
SARTORIUS
RECTUS FEMORIS (cut)
VASTUS LATERALIS
VASTUS INTERMEDIUS
VASTUS MEDIALIS
Iliotibial tract
RECTUS FEMORIS (cut)
Section of fascia lata (cut)
Tendon of quadriceps femoris
Patellar ligament
PSOAS MAJOR
Sacrum
Inguinal ligament
PECTINEUS
ADDUCTOR LONGUS
GRACILIS
ADDUCTOR MAGNUS
Patella
(a) Anterior superficial view
Iliac crest
GLUTEUS MAXIMUS (cut)
Sacrum
Coccyx
Sciatic nerve
GRACILIS
SARTORIUS
GLUTEUS MEDIUS (cut)
GLUTEUS MINIMUS
PIRIFORMIS
GLUTEUS MAXIMUS (cut)
ADDUCTOR MAGNUS
SEMITENDINOSUS
BICEPS FEMORIS
SEMIMEMBRANOSUS
Vastus lateralis
(b) Posterior superficial view

Hip: Function – Range of Motion

- flexion: 120˚
- extension: 20˚
- internal rotation: 30˚
- external rotation: 60˚
- abduction: 45˚
- adduction: 30˚

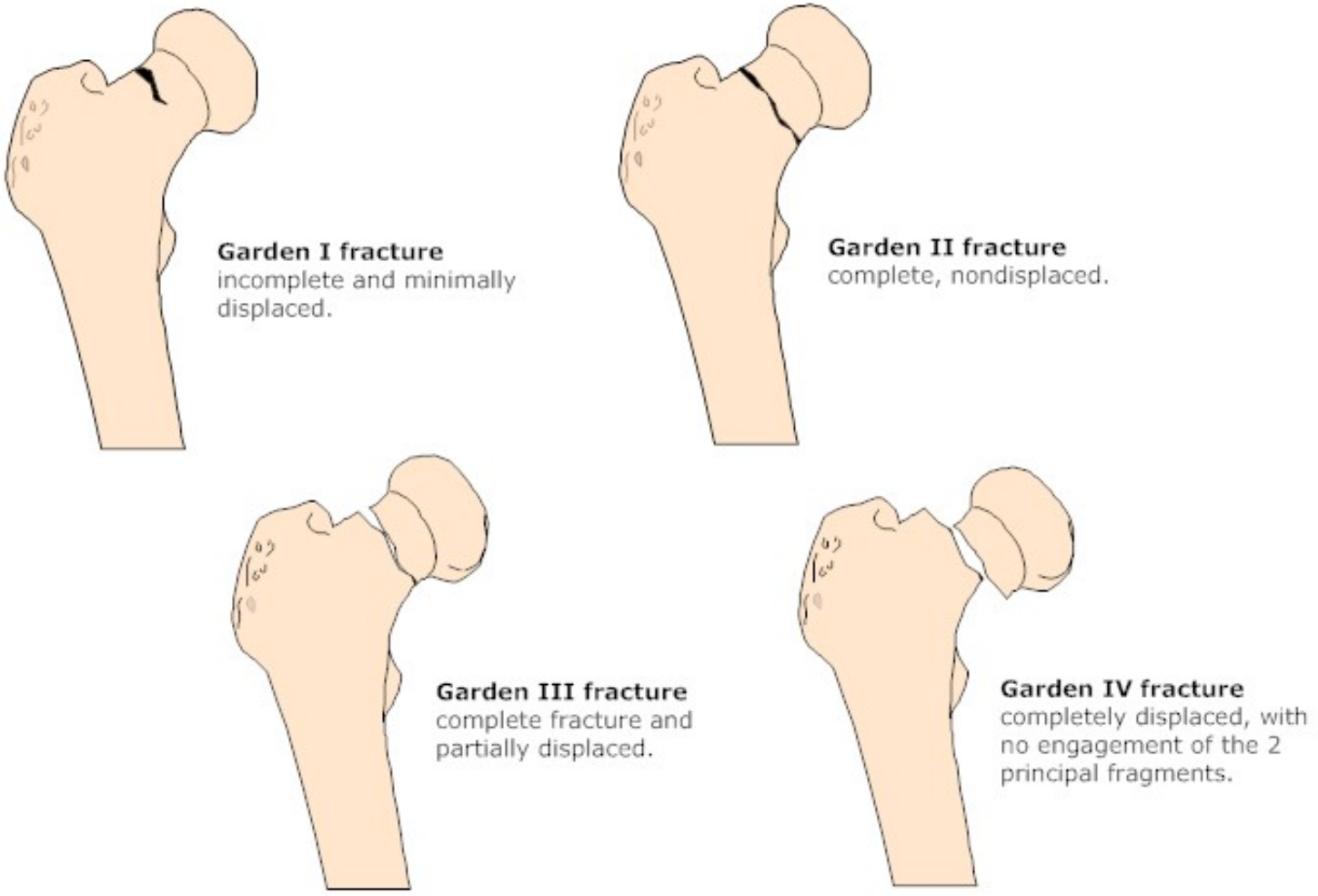
Hip: Function – General

- transfers load from trunk to lower extremity (and vice versa)
- maintains balance (M/L)
- permits stable rotational movement in 3 planes (circumduction)
- absorbs shock

Hip: Pathology - Fractures

- primarily caused by:
  - high energy impacts
    - fall from height
    - automobile crashes
- osteoporosis
  - chicken v. egg
  - fall then fracture, not fracture then fall

Hip: Pathology - Fractures



LifeART Collection Images Copyright © 1989-2001 by Lippincott Williams & Wilkins, Baltimore, MD

Hip: Pathology - Fractures

AO Classification of Intertrochanteric Fractures

A1: Simple (2-fragment) fractures

A1.1 Fractures along the intertrochanteric line

A1.2 Fractures through the greater trochanter

A1.3 Fractures below the lesser trochanter

A2: Multifragmentary fractures

A2.1 With one intermediate fragment

A2.2 With 2 intermediate fragments

A2.3 With more than 2 intermediate fragments

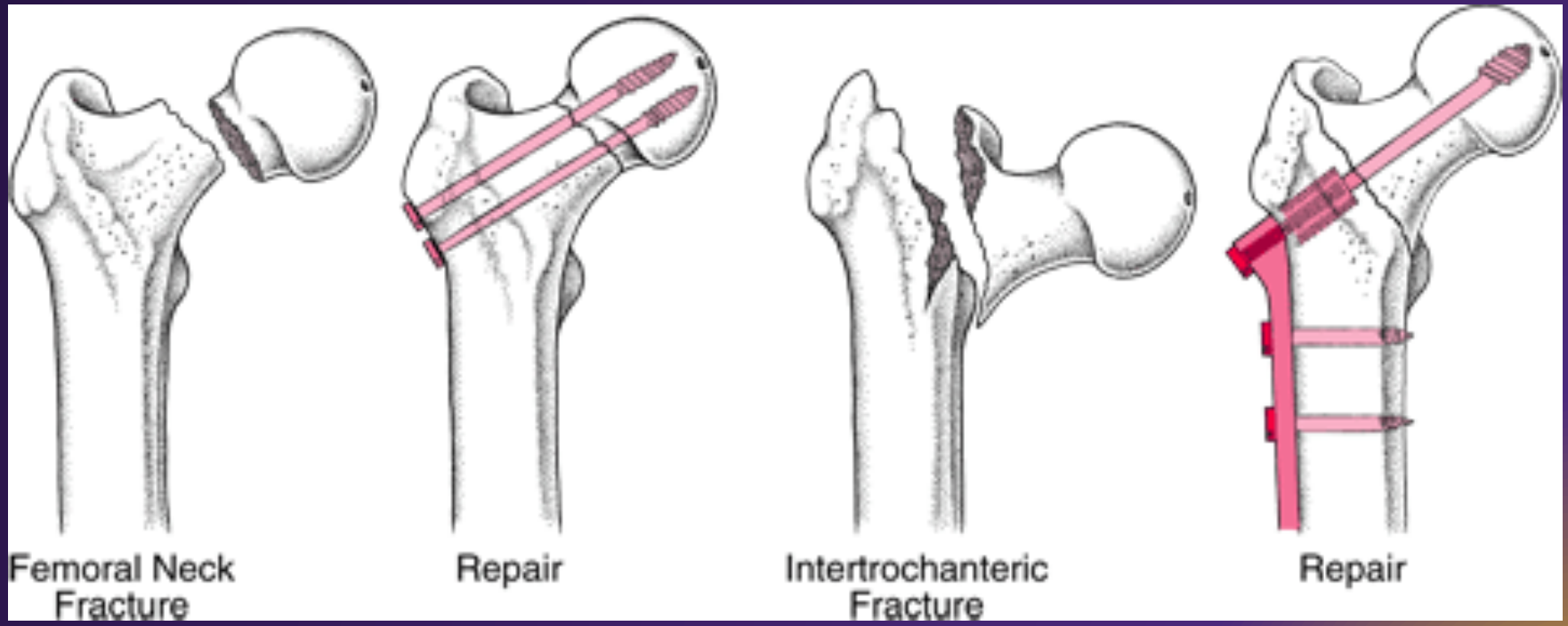
A3: Intertrochanteric fractures

A3.1 Simple, oblique

A3.2 Simple, transverse

A3.3 With a medial fragment

Hip: Pathology – Fracture Treatment



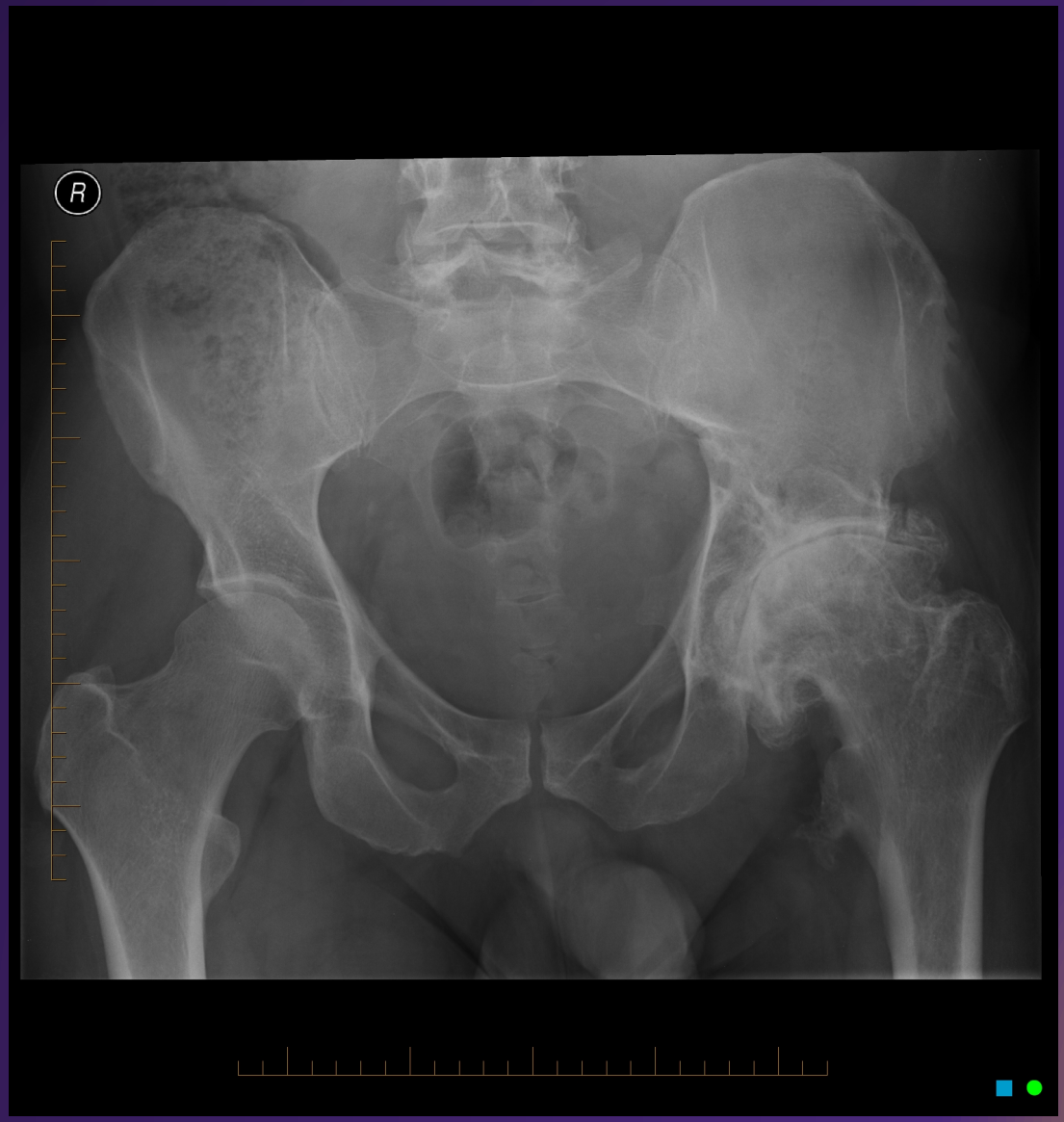
Cannulated Bone Screws

Compression screw and side plate

Hip: Pathology – Arthroplasty

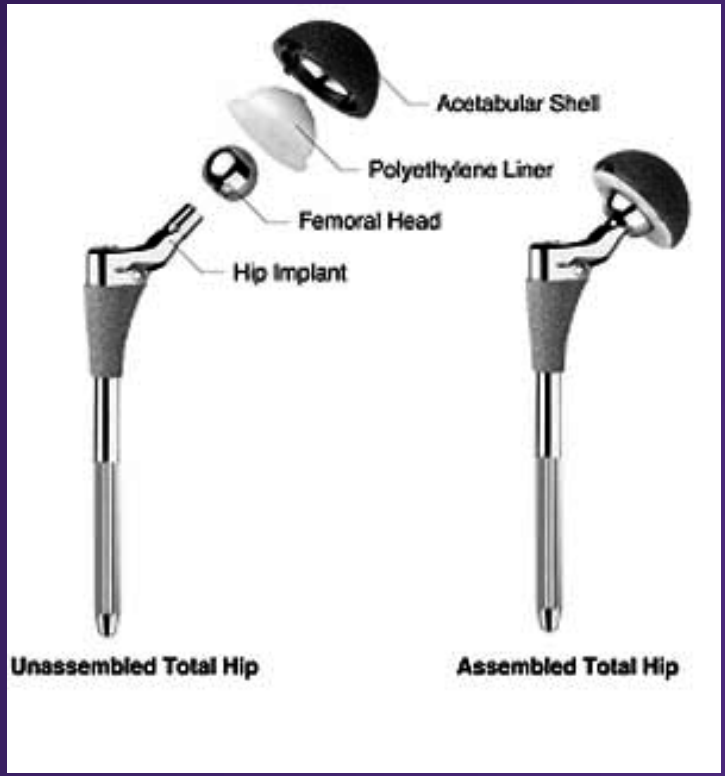
- indications:
  - end stage osteoarthritis (vast majority)
  - hip fracture with interrupted vasculature
  - rheumatoid arthritis
  - congenital deformities
- 332,000 per year in US (National Hospital Discharge Survey: 2010)
- forces in hip:
  - 3x to 4x BW in walking
  - 5x jogging or on stairs
  - 8x stumbling

Hip: Pathology – Arthroplasty



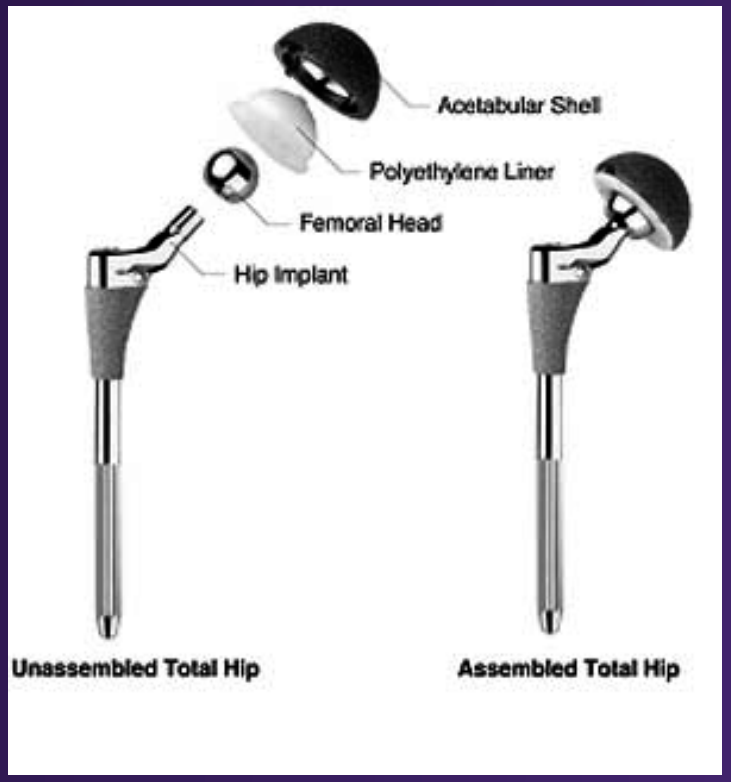
Hip: Pathology – Arthroplasty

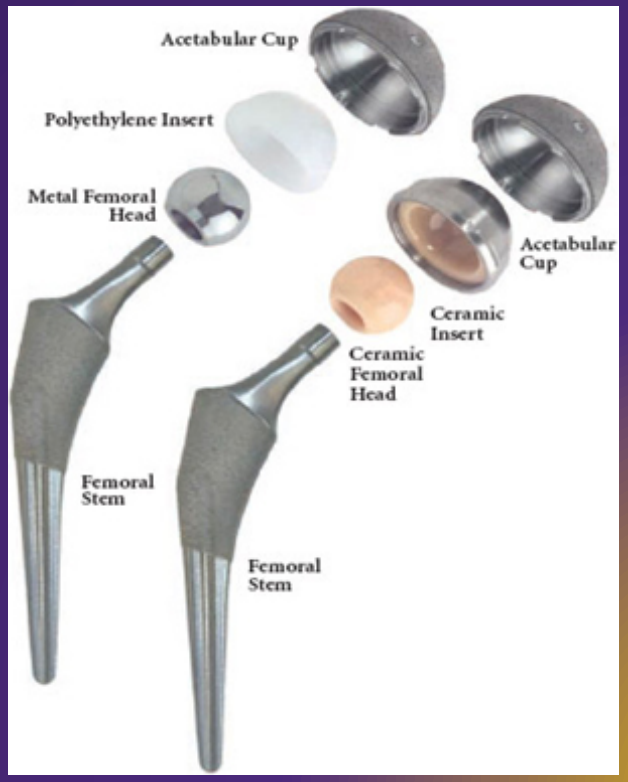
- Components:
  - femoral component (stem/ball)
  - acetabular component (cup: shell /liner)



Hip: Pathology – Arthroplasty

- materials:
  - metal and polyethylene
  - ceramic and ceramic, metal and metal





Hip: Pathology – Arthroplasty

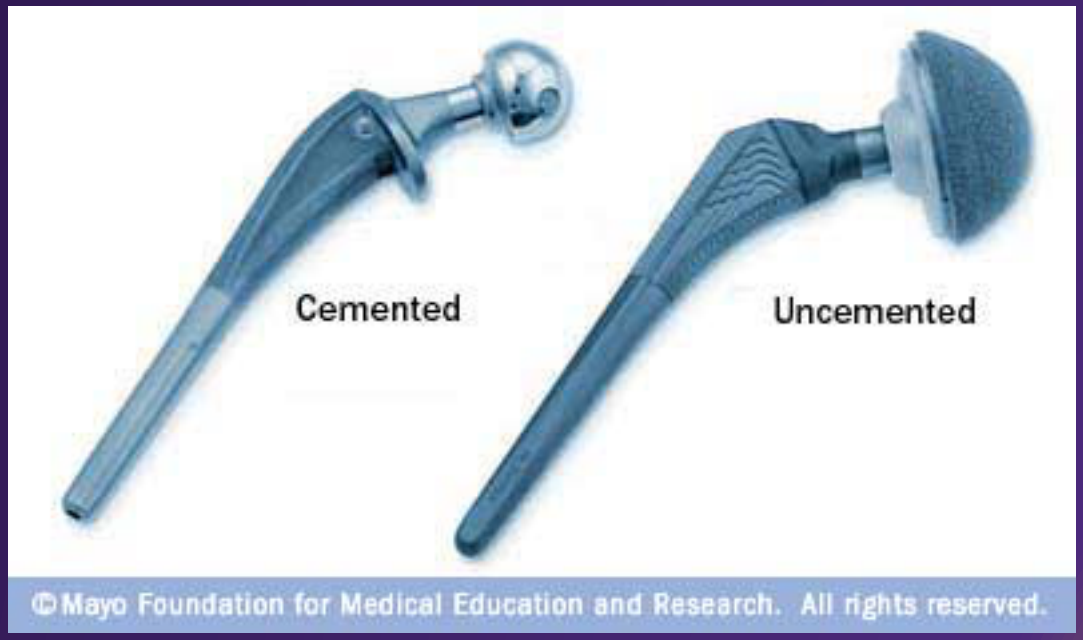
- metal and metal – submicron metallic debris
  - hip and articular surface replacement (ASR)
- 11/2013 – J&J agreed to a ~$4B settlement
- http://tinyurl.com/kzlkxm7

Hip: Pathology – Arthroplasty

- increase ion levels (chromium and cobalt)
- metallosis (build up of metals)
- hypersensitivity to metal
- osteolysis and early implant failure

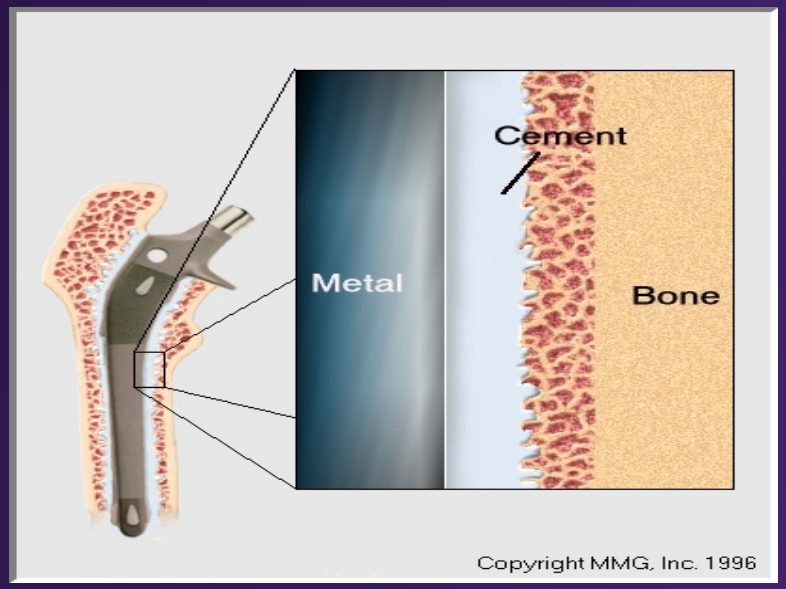
Hip: Pathology – Arthroplasty

- component attachment

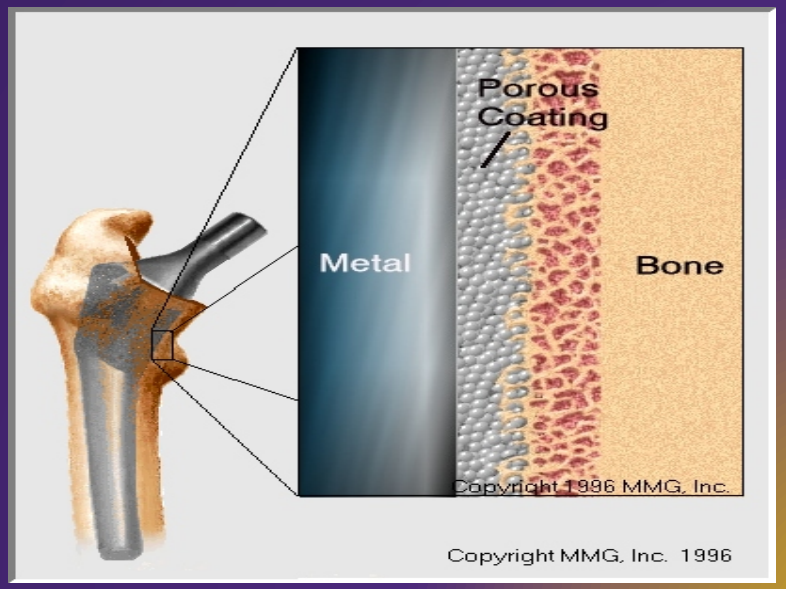


Hip: Pathology – Arthroplasty

- cemented advantages:
  - used osteoporotic bone
  - add antibiotic, reduced risk of post-surgical infection
  - dries within 10 minutes, confident it is firmly in place



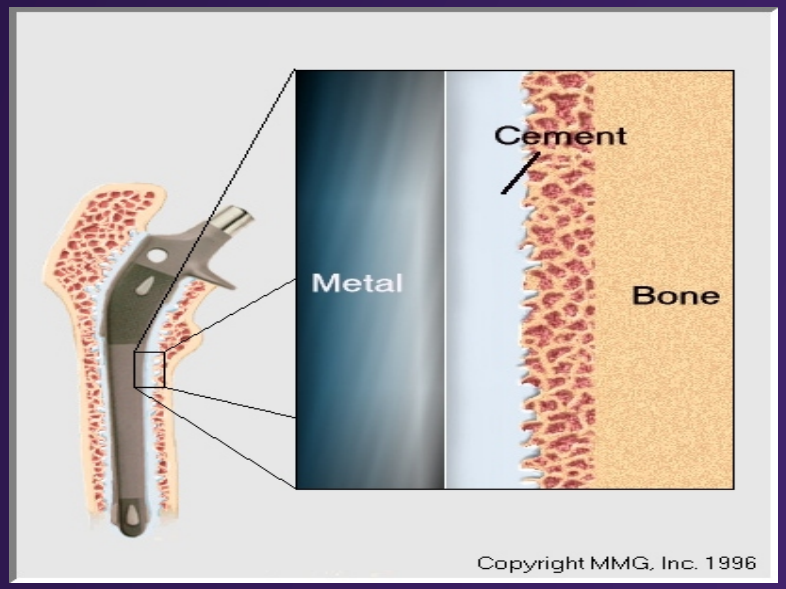
cemented



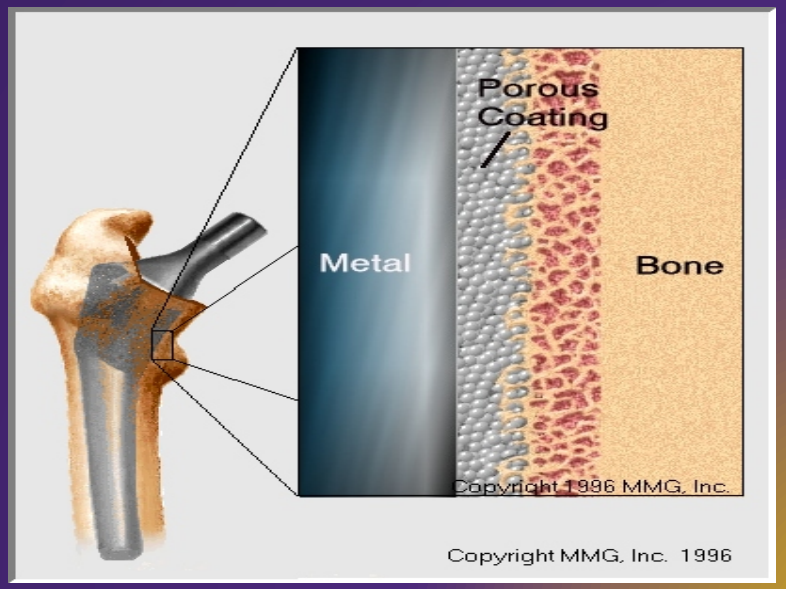
cementless

Hip: Pathology – Arthroplasty

- cemented disadvantages:
  - cement breakdown can lead to loosening, revision
  - cement debris can irritate surrounding tissue
  - rare, but cement can enter blood stream and end up in the lungs – life threatening



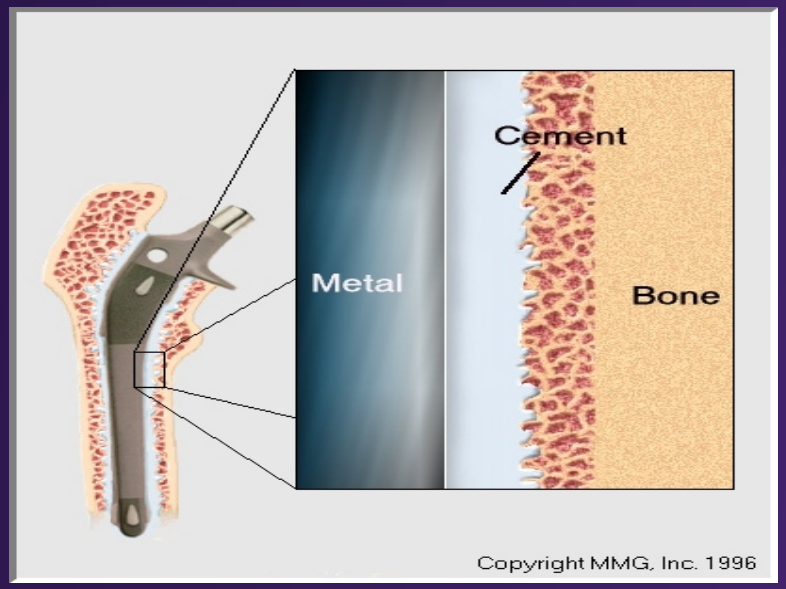
cemented



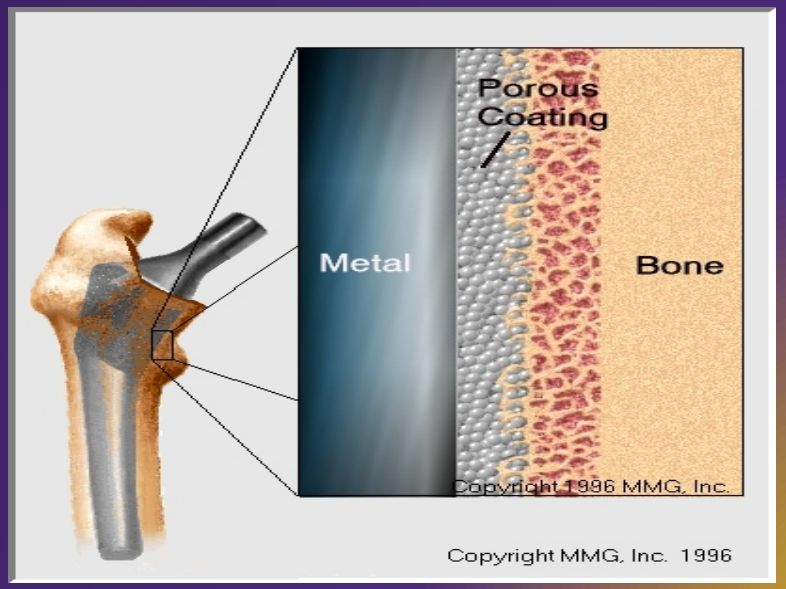
cementless

Hip: Pathology – Arthroplasty

- cementless advantages:
  - better long-term bond between bone and prosthesis
  - no worry about potential breakdown of cement



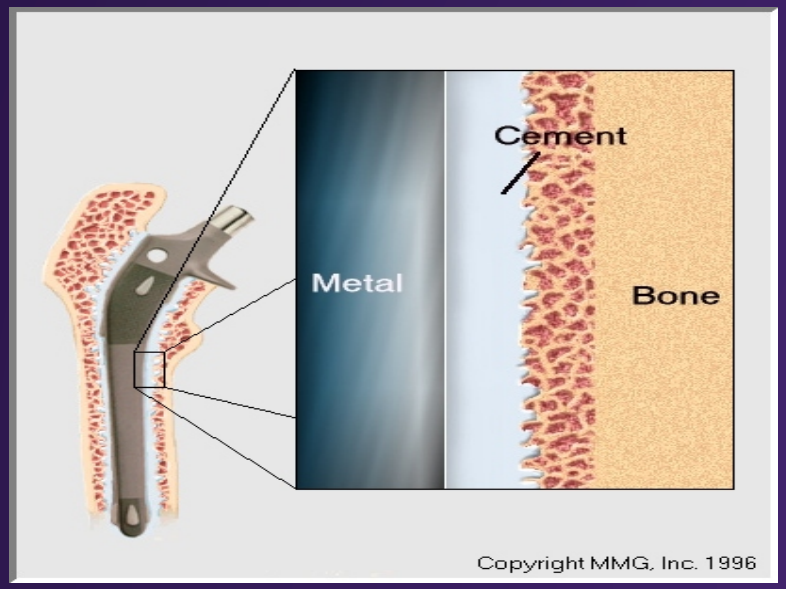
cemented



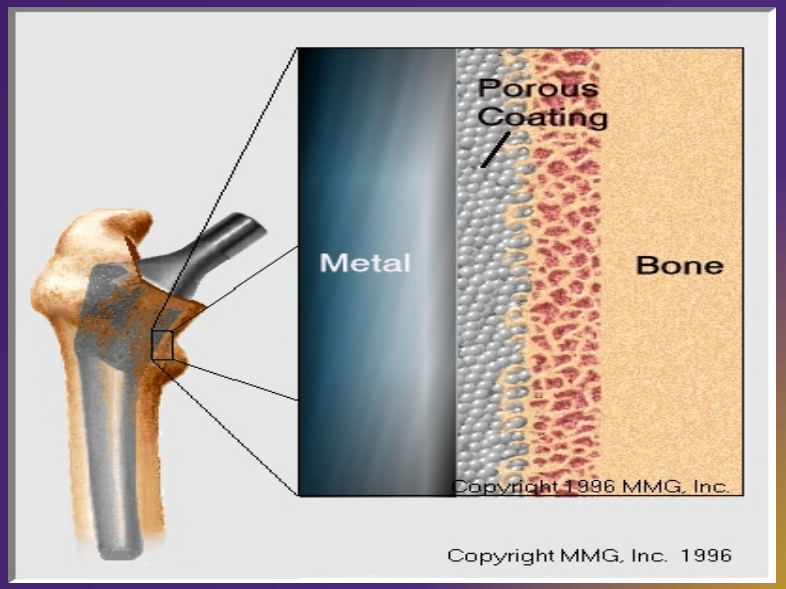
cementless

Hip: Pathology – Arthroplasty

- cementless disadvantages:
  - require healthy bones
  - patients with low density (osteoporotic) are ineligible
  - up to 3 months for bone to grow into joint component



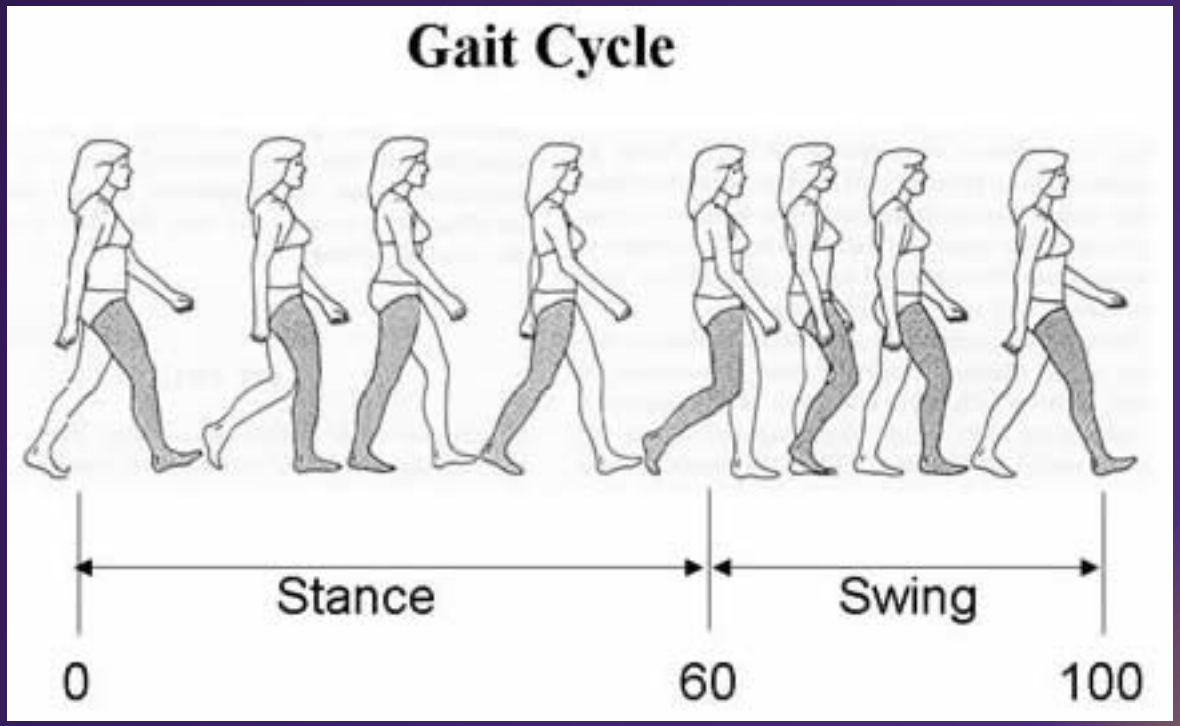
cemented



cementless

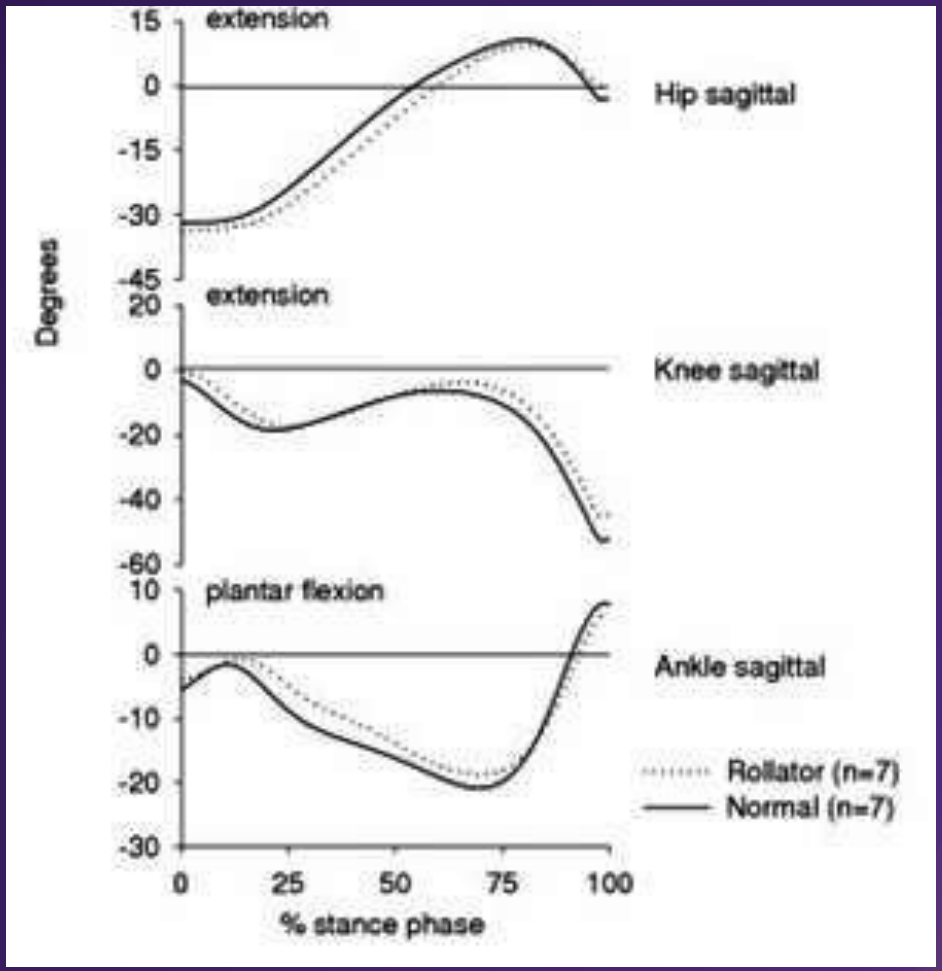
Hip Function

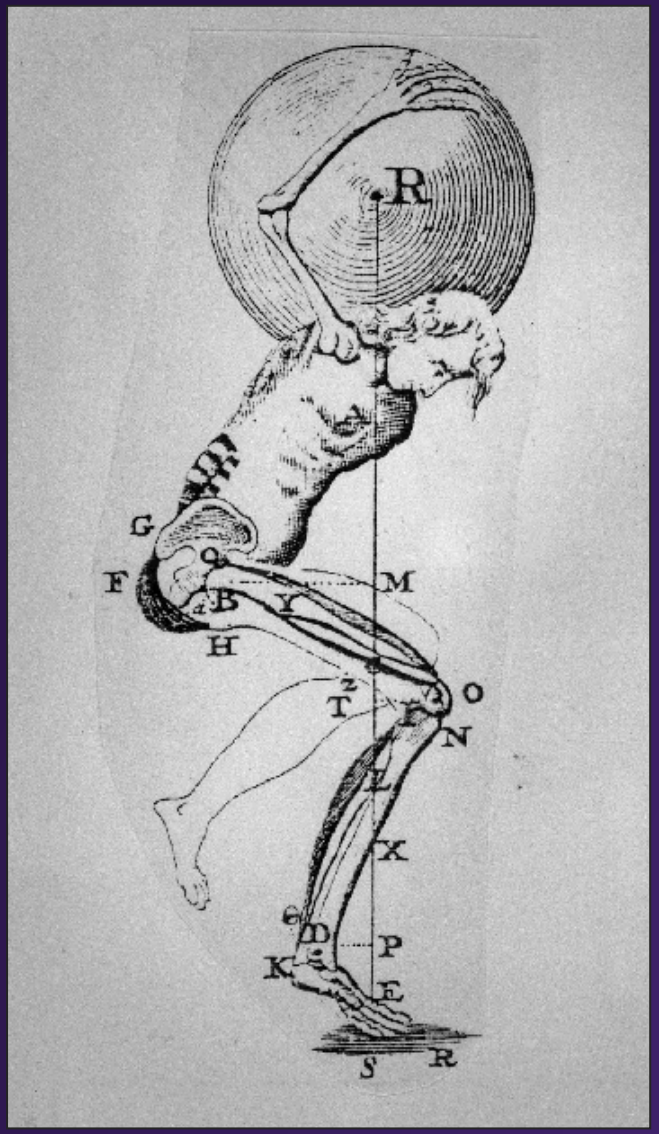
- motion data



Hip Function

- motion data





Musculoskeletal Biomechanics

BIOEN 520 | ME 527

Session 14A2

Biomechanics of the Knee

Session 14A1 Review…

- Hip
- Anatomy
- General function
- Pathology

Session 14A2 Overview…

- Knee
- Anatomy
- General function
- Pathology

Knee: Anatomy

- type of joint:
  - synovial (diarthrodial)
  - hinge
  - 2 joints
    - tibiofemoral
    - patellofemoral

Knee: Anatomy

- bones:
  - femur, tibia, patella, fibula*

Knee: Anatomy

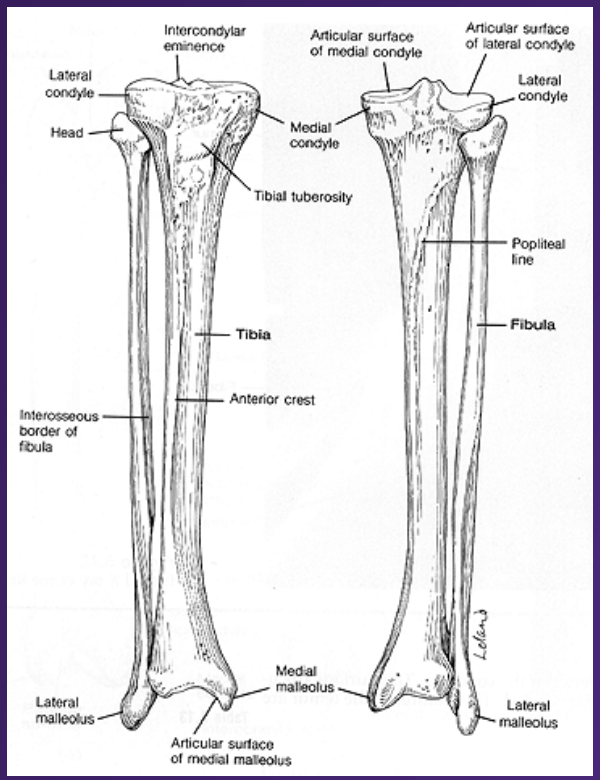
- femur:
  - distal femur - medial and lateral condyle, anterior patellar groove

Knee: Anatomy

- tibia:
  - proximal tibia - medial and lateral condyle, or tibia plateau
  - intercondylar eminence

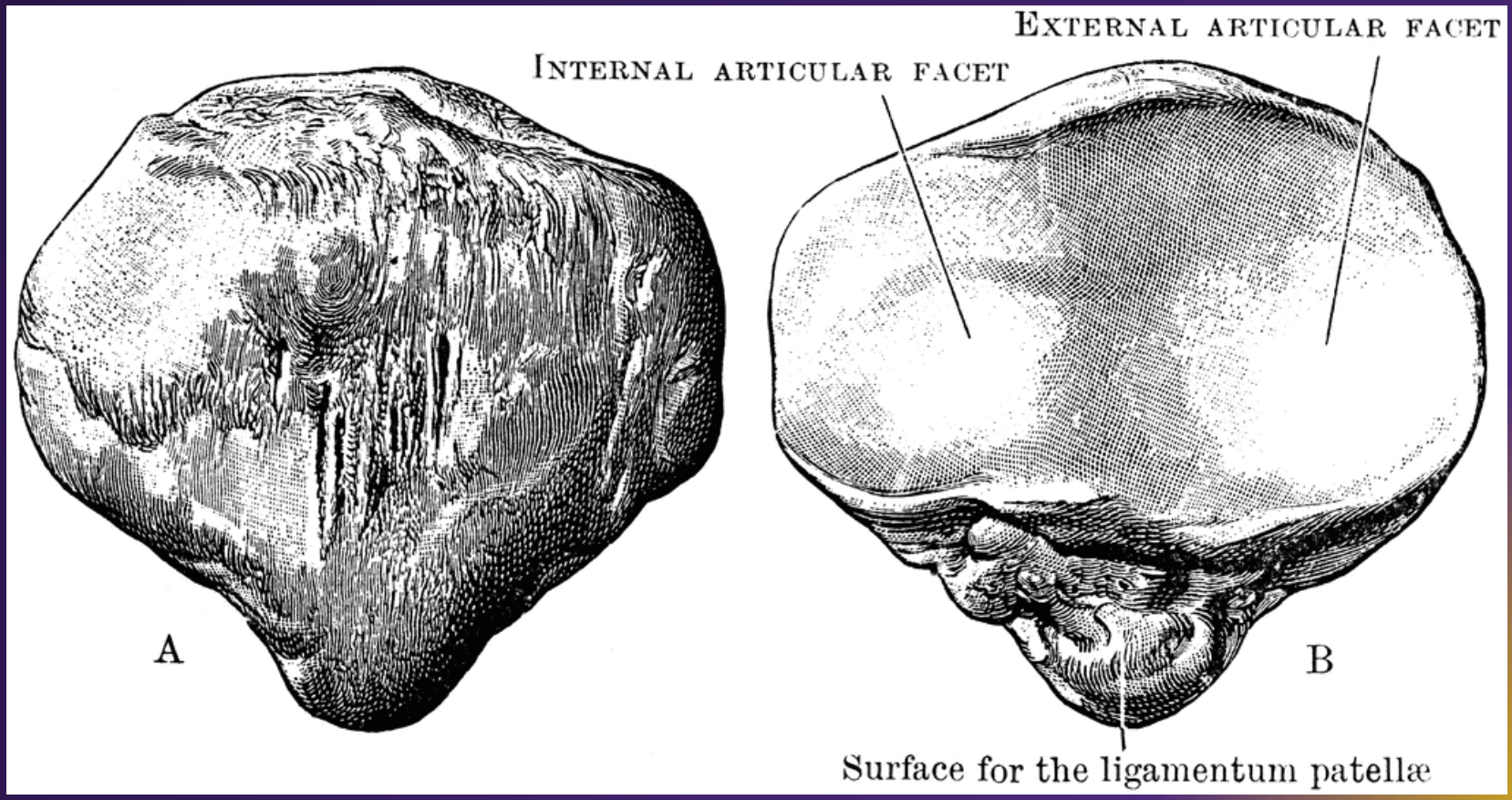
Knee: Anatomy

- tibia:
  - second longest bone in body
  - stronger than fibula, about 85% of BW



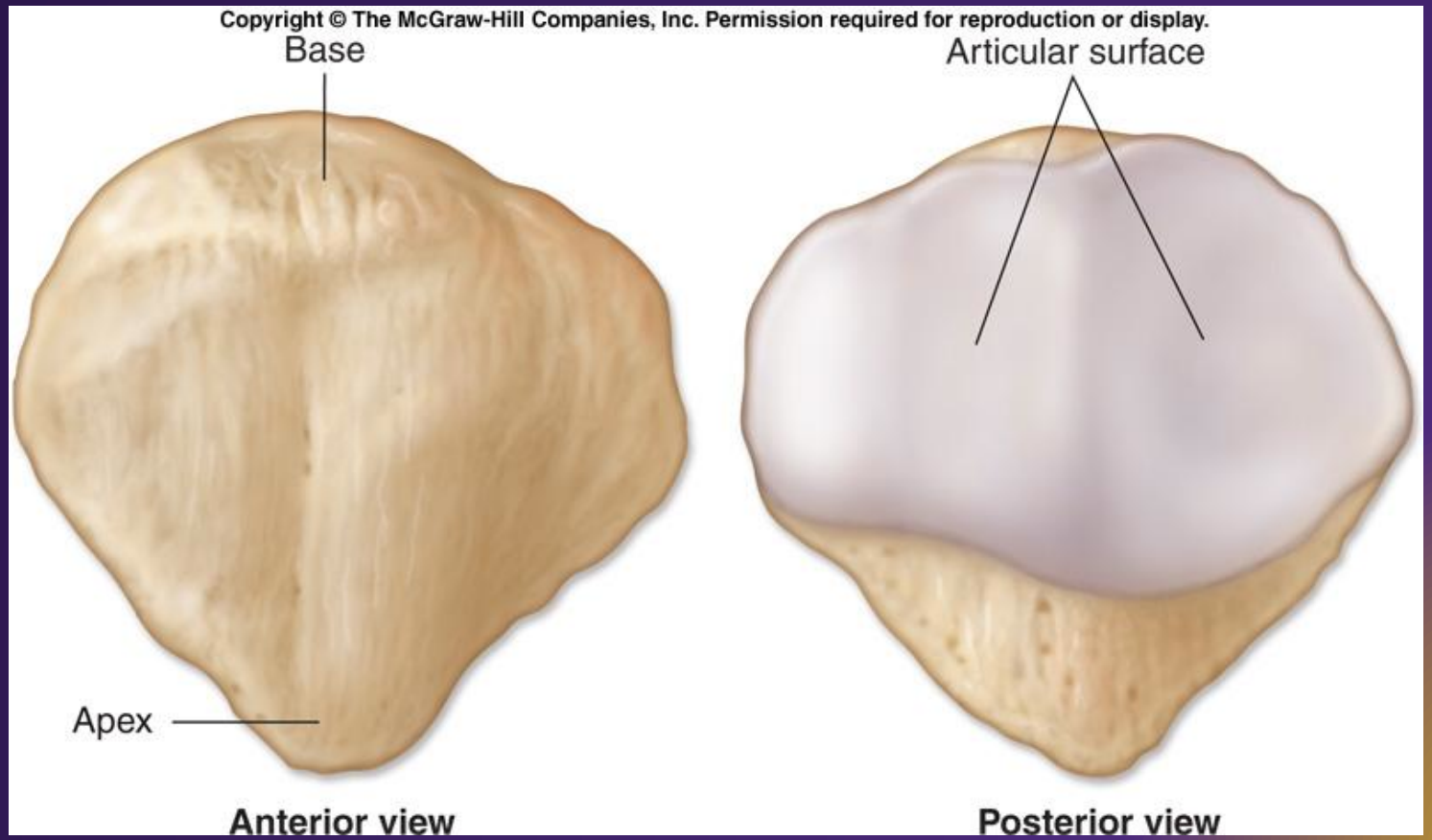
Knee: Anatomy

- •patella:
  - posterior face articulates, two facets



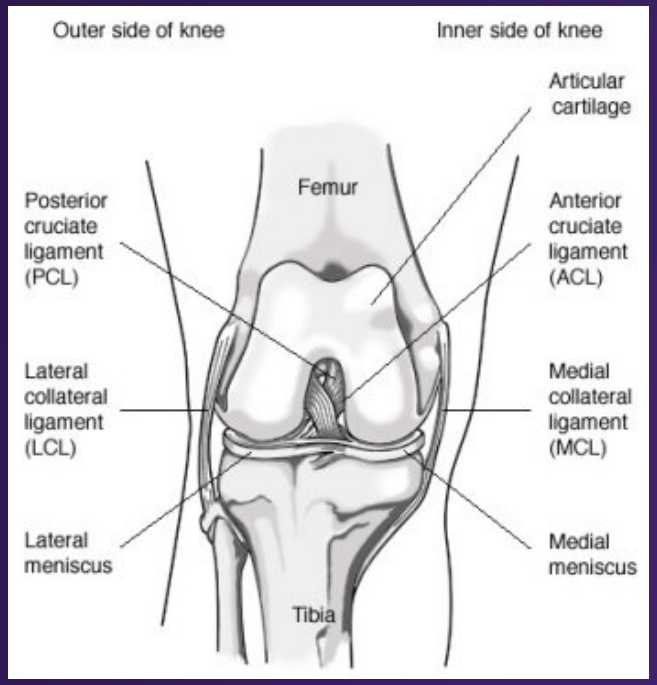
Knee: Anatomy

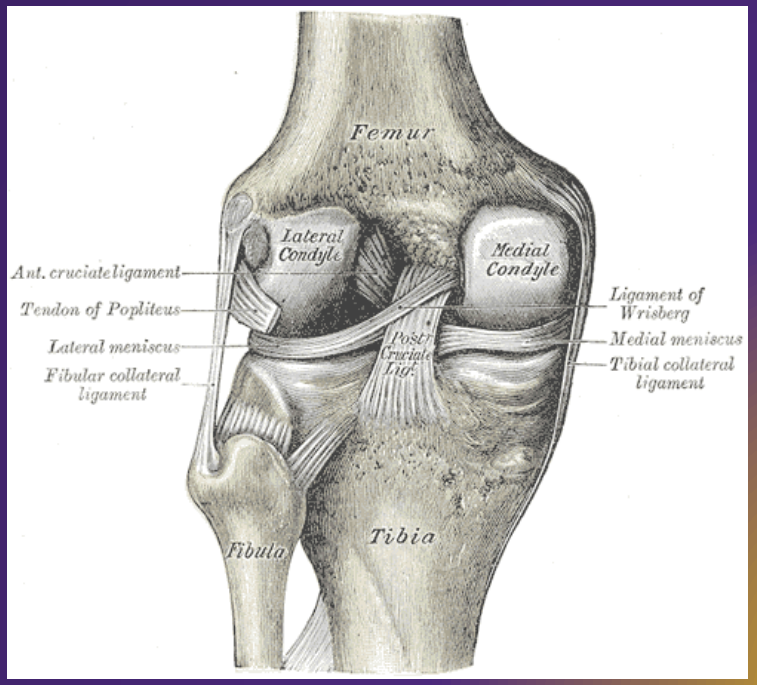
- patella:
  - largest sesamoid in the body



Knee: Anatomy

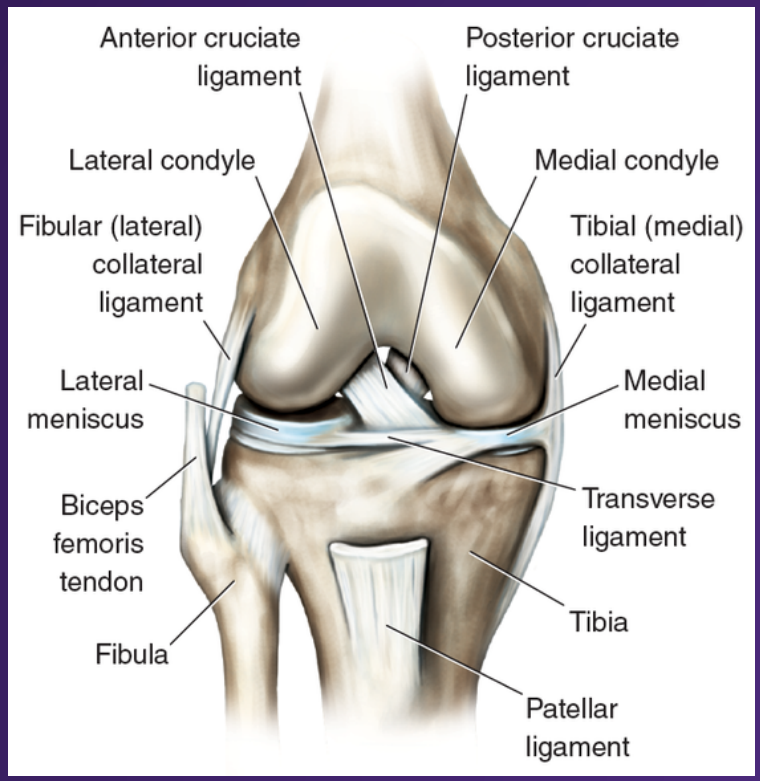
- ligaments:
  - anterior and posterior cruciate (ACL and PCL)
  - limits anterior and posterior tibial displacement
  - limits internal/external tibial rotation





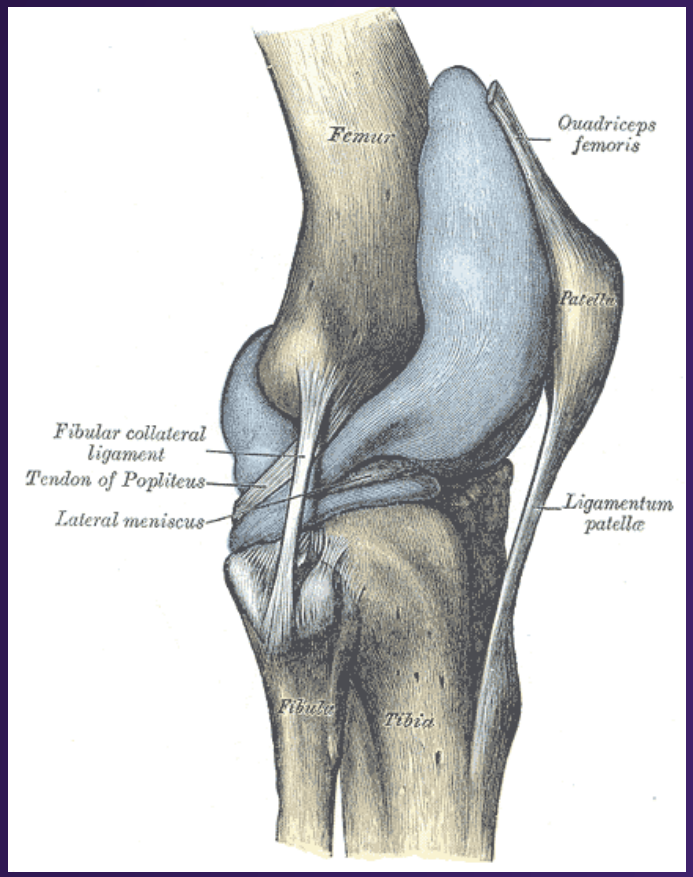
Knee: Anatomy

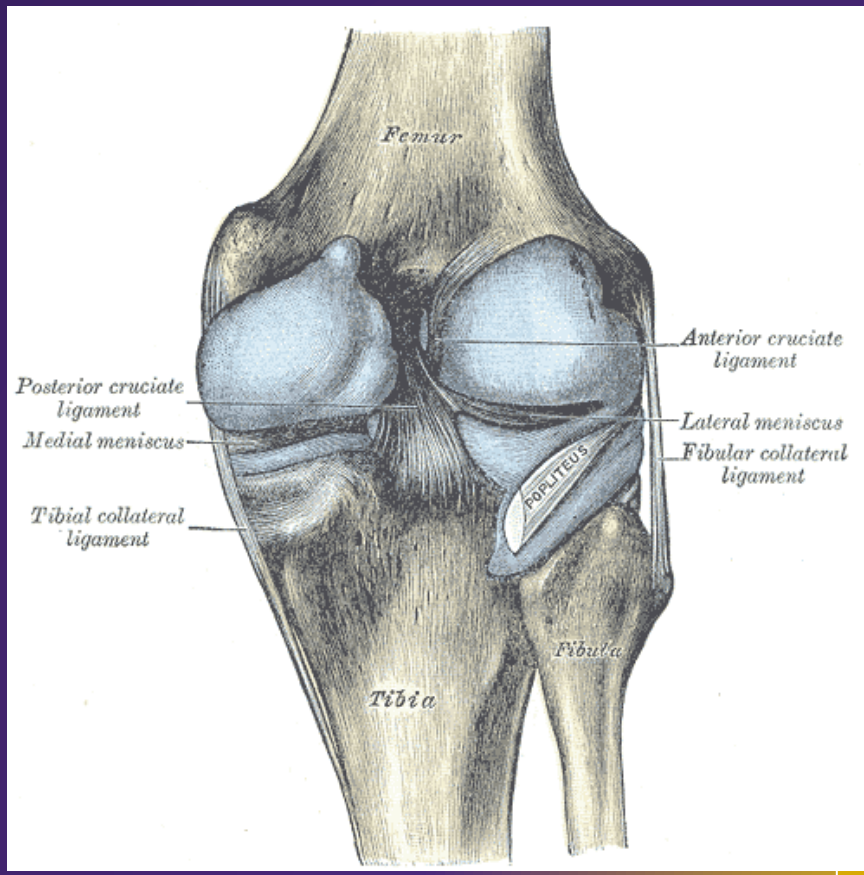
- ligaments:
  - medial and lateral collaterial (MCL and LCL)
  - limits varus/valgus knee motion



Knee: Anatomy

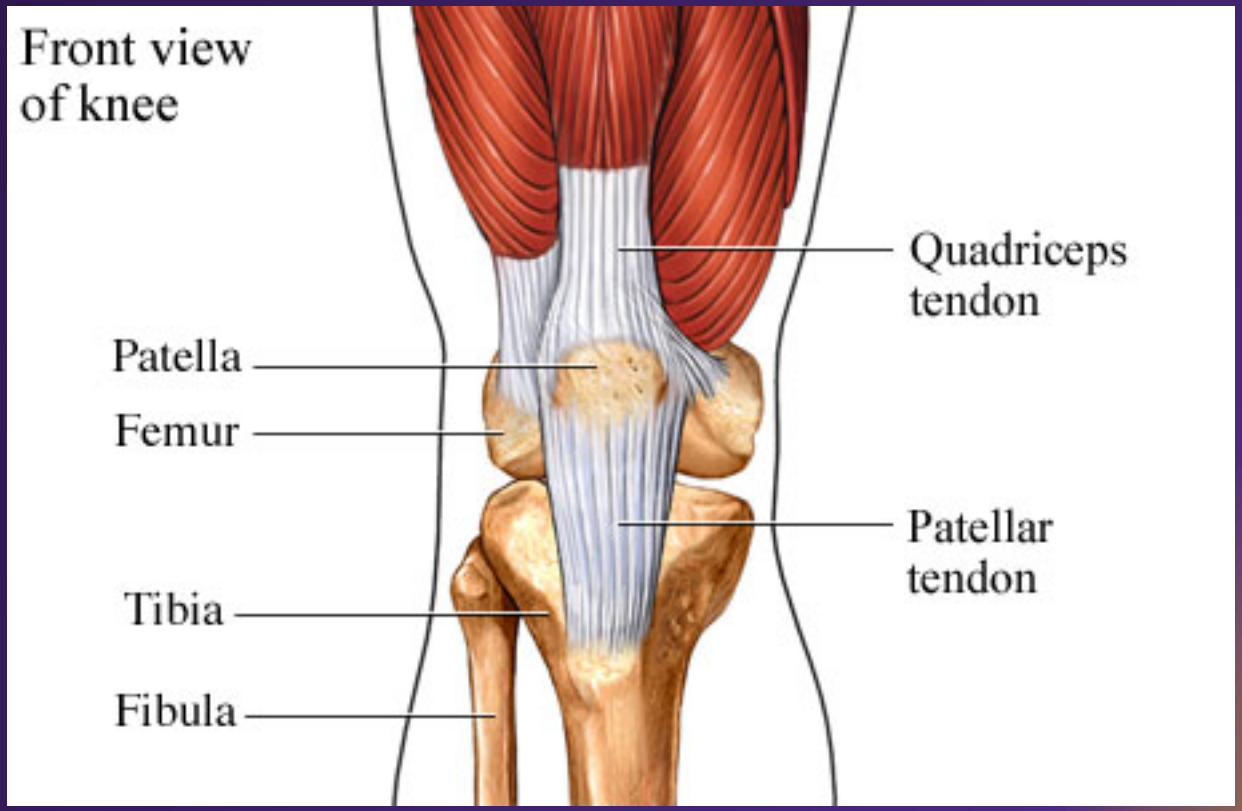
- capsule: not as fibrous as the hip
  - anatomy is quite complex (Fig 5, 6, 7, 8, 9)





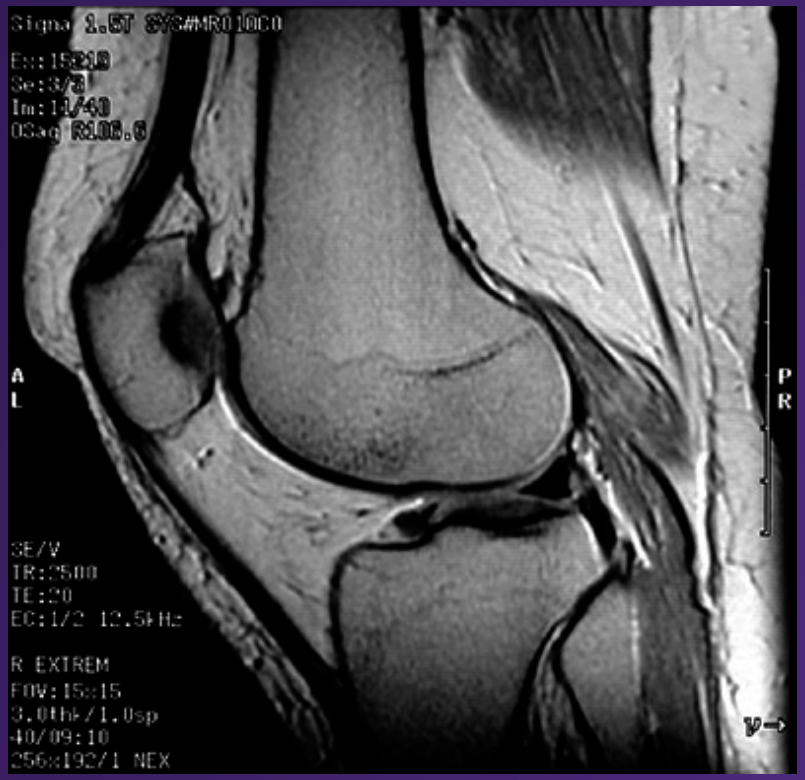
Knee: Anatomy

- patellar tendon
  - ligament vs. tendon



Knee: Anatomy

- infrapatellar fat pad (Hoffa's fat pad)
  - cushions patella
  - can become impinged

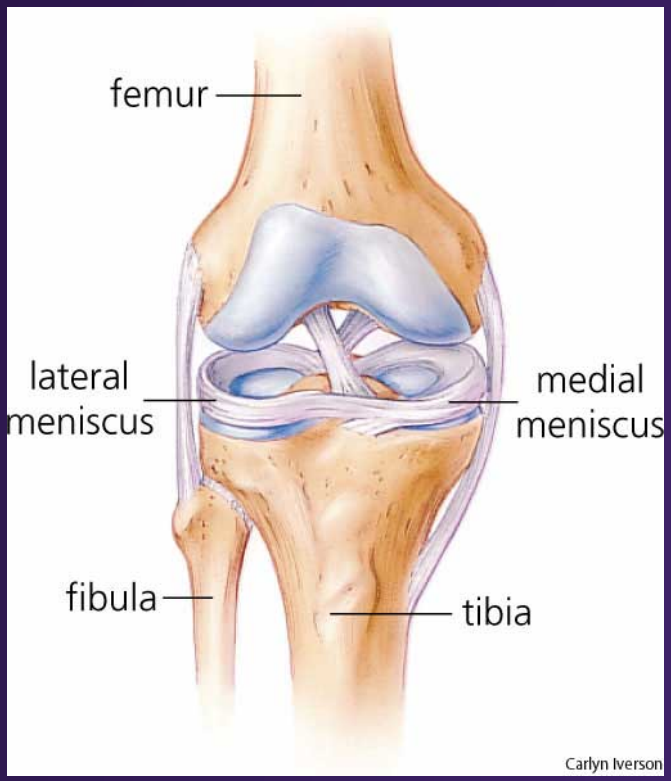


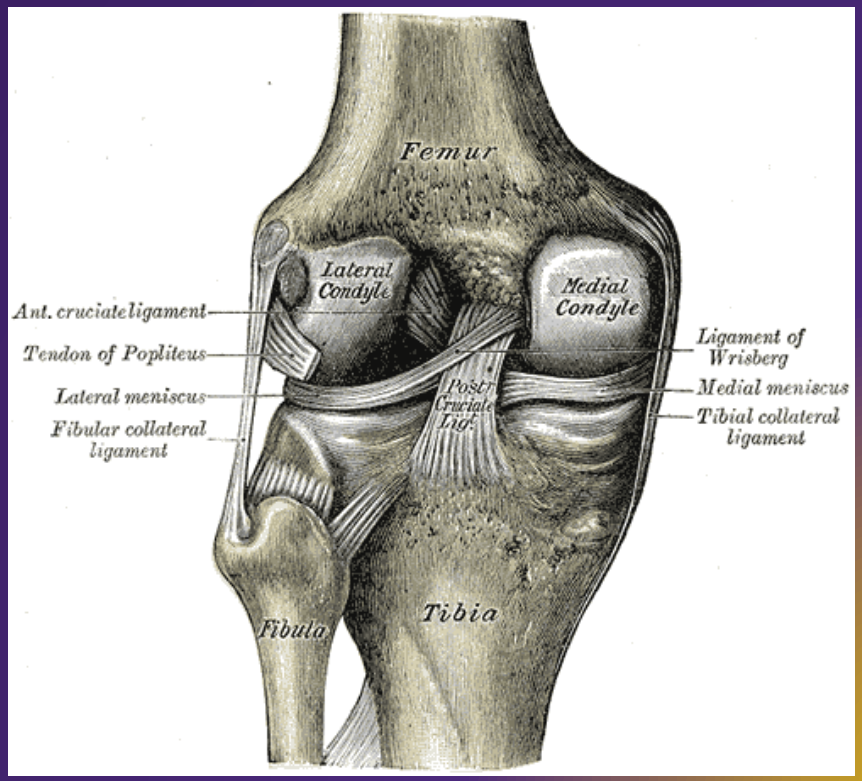
Knee: Anatomy

- cartilage:
  - all articular surfaces femur, tibia, patella

Knee: Anatomy

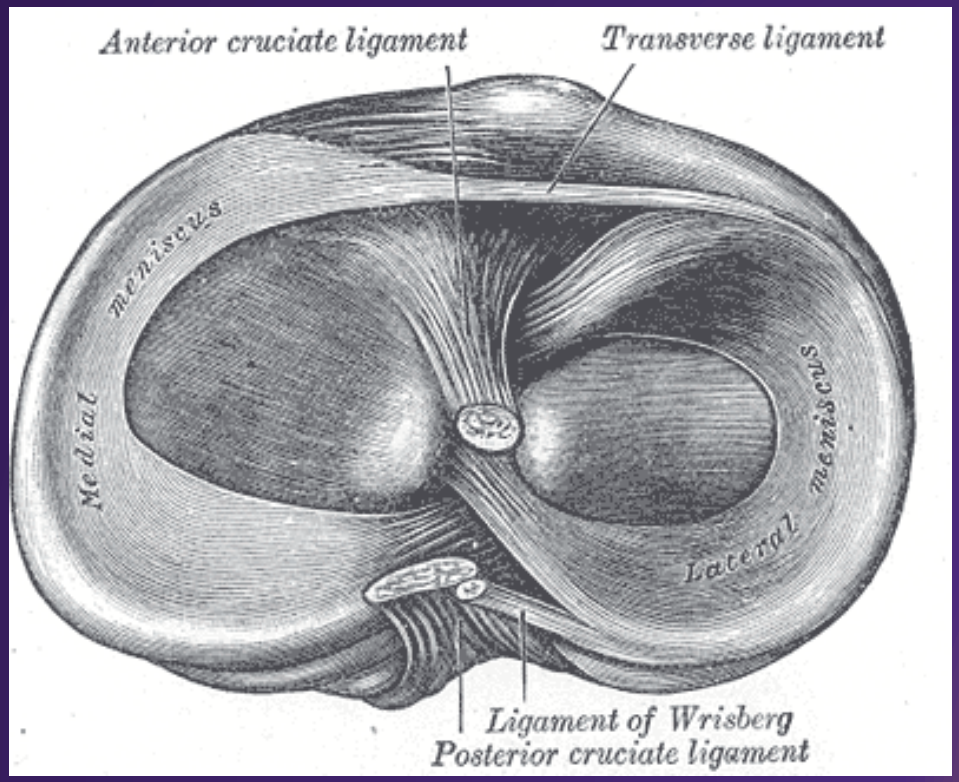
- cartilage:
  - medial and lateral meniscus
  - fibrocartilage, cushioning



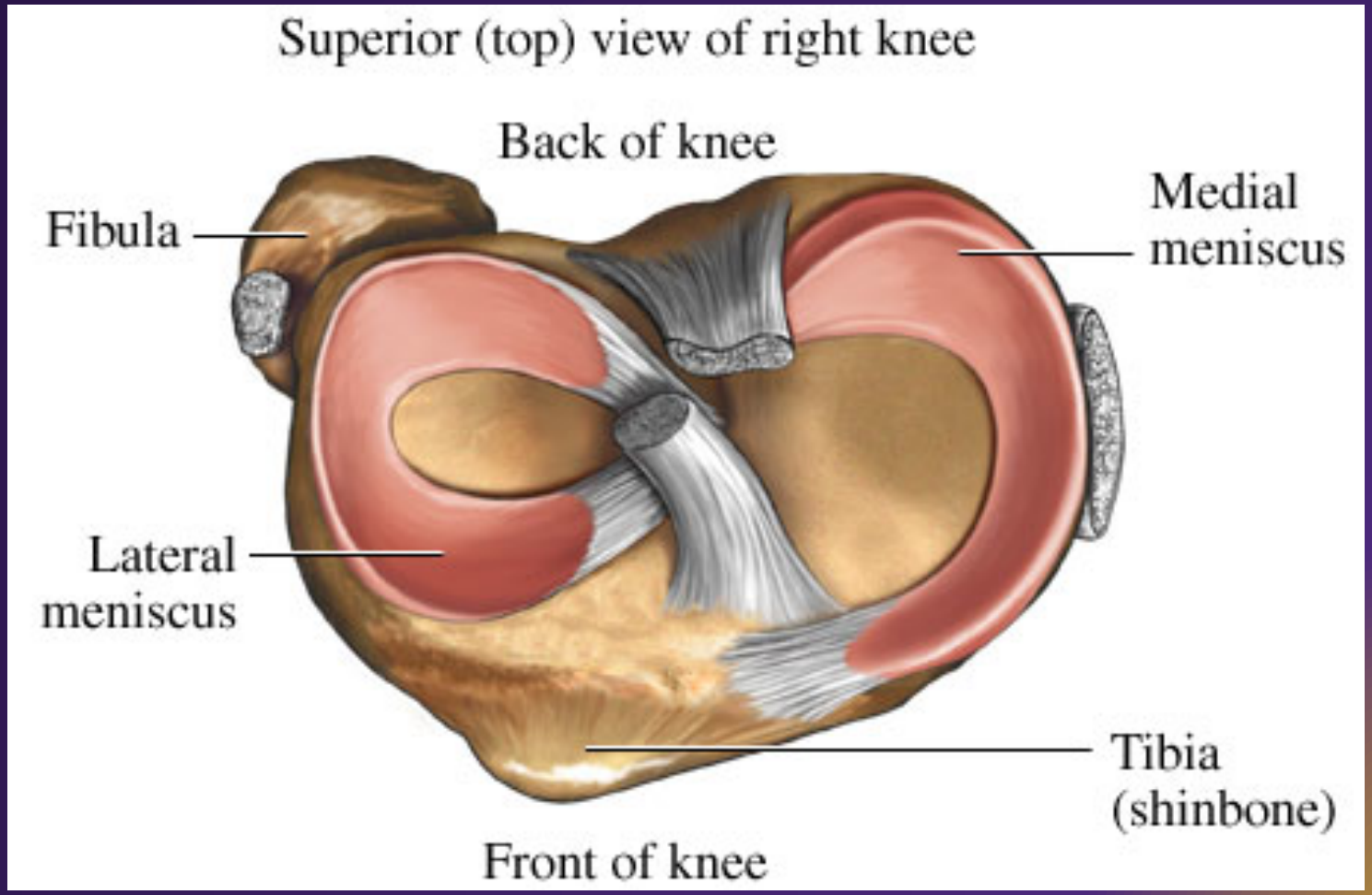


Knee: Anatomy

- cartilage:
  - medial and lateral meniscus
  - fibrocartilage, cushioning

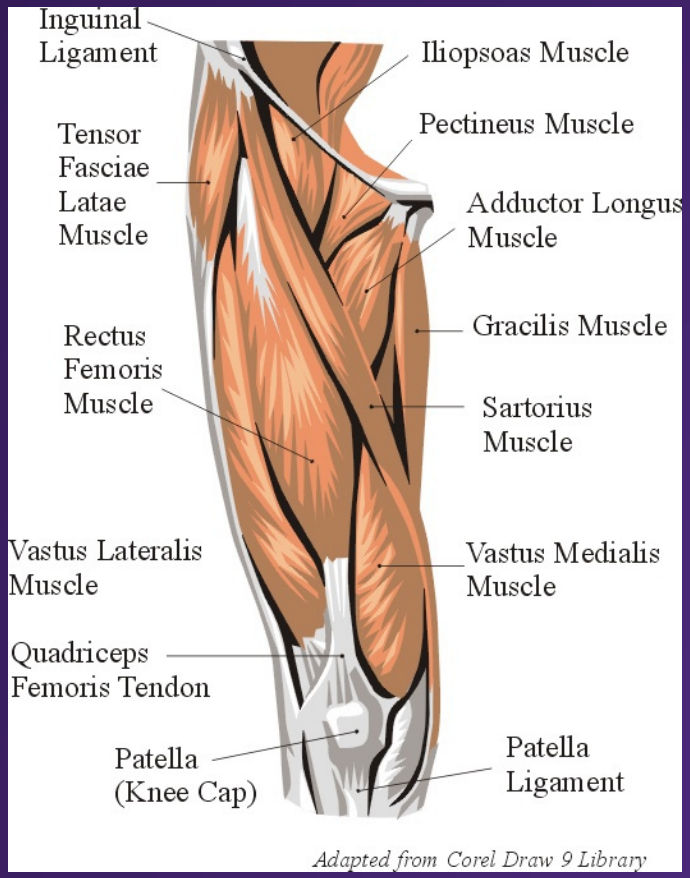


Knee: Anatomy



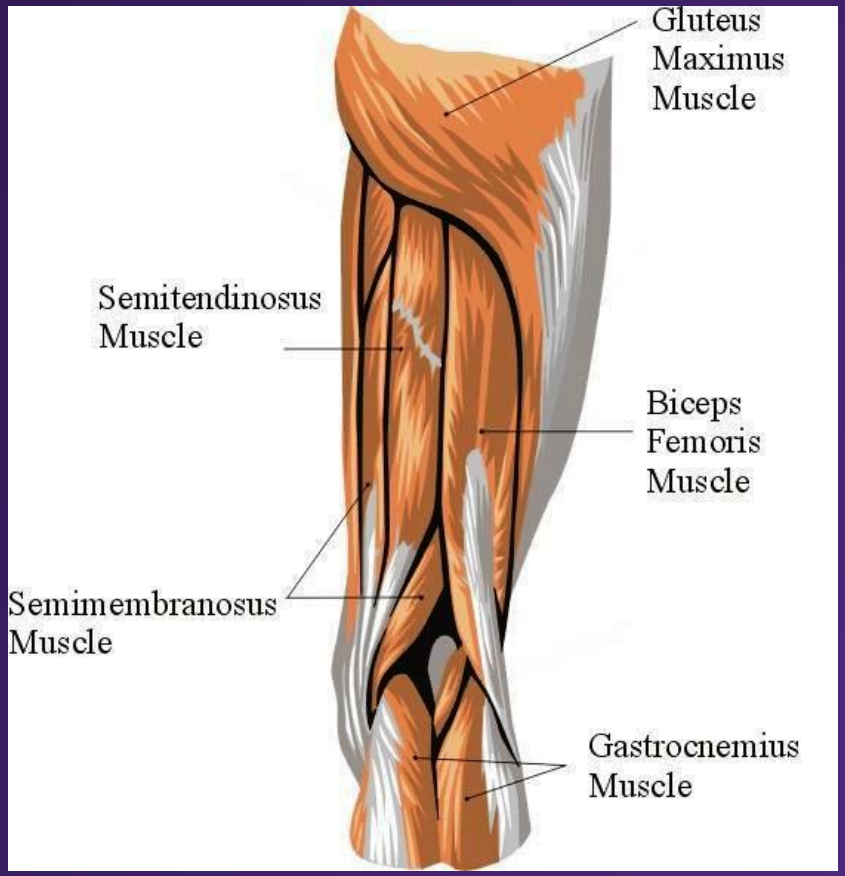
Knee: Anatomy

- muscles: extensors
  - quadriceps

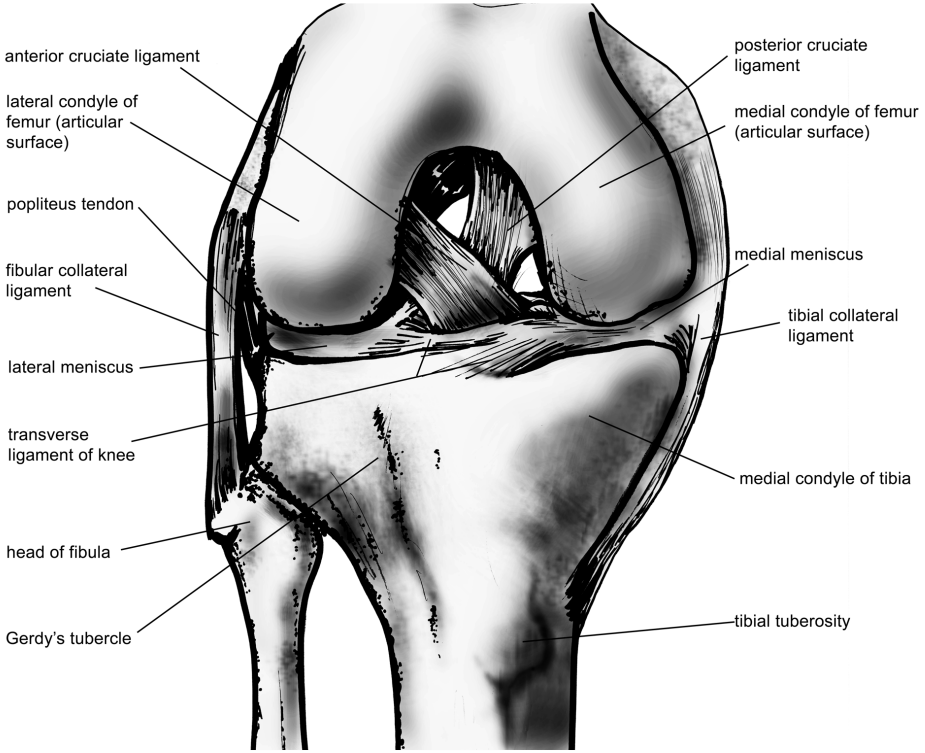


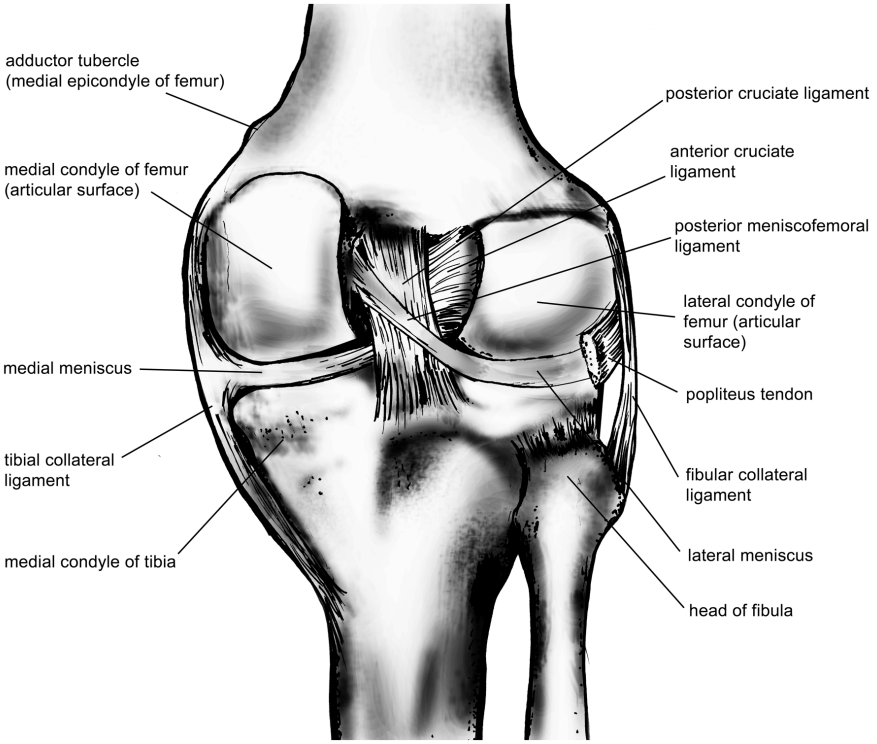
Knee: Anatomy

- muscles: flexors
  - hamstrings



Knee: Anatomy





Knee: Function – Range of Motion

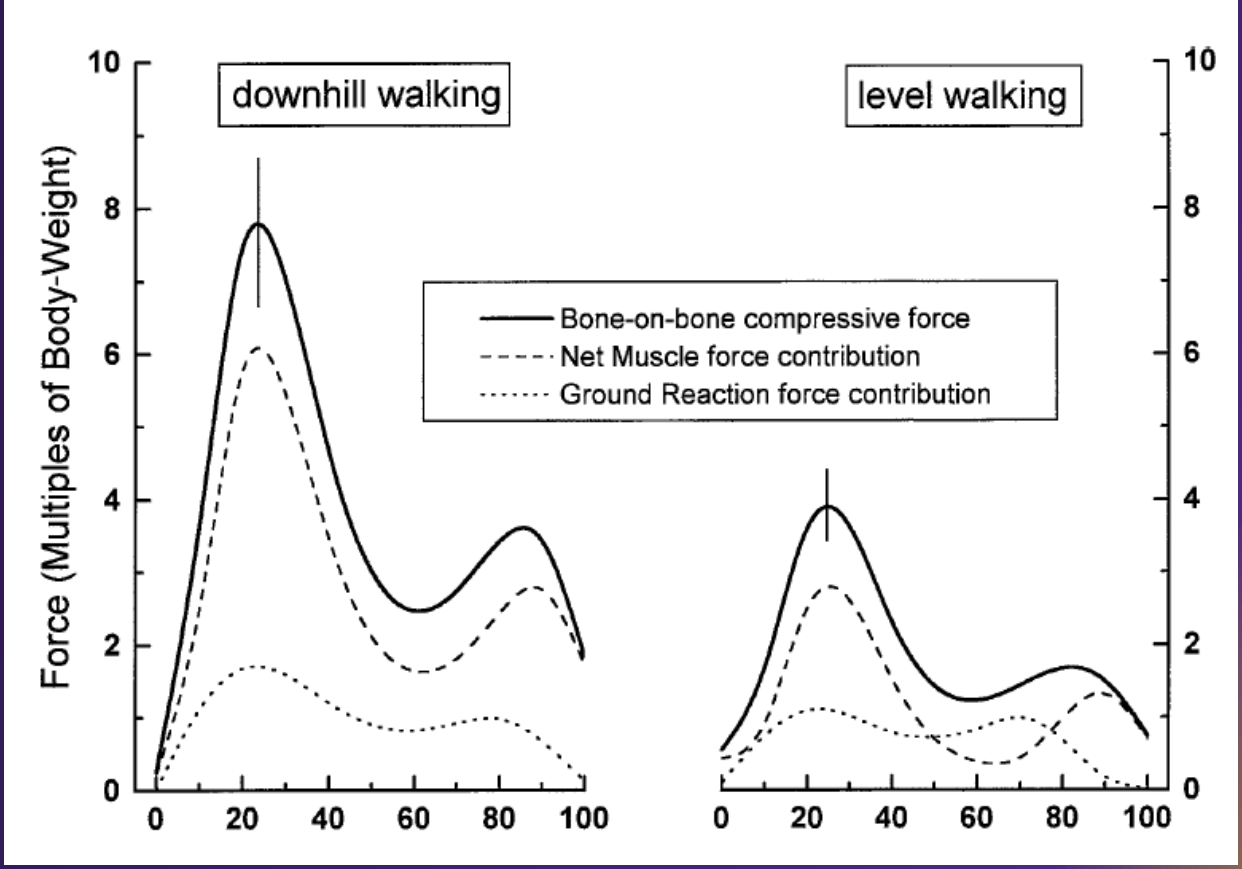
- flexion: 160˚
- extension: 0˚
- internal rotation: 0˚ or 30˚ with knee flexed
- external rotation: 0˚ or 45˚ with knee flexed
- abduction: 0˚
- adduction: 0˚

Knee: Function – General

- transfers load from thigh to lower leg
- allows distance between trunk and ground to be varied; works essentially by axial compression under gravity
- allows long axis rotation when knee is flexed
- great stability in full extension
  - injuries are fractures of articular surface or ligament rupture
- great mobility after some flexion
  - knee unstable - ligaments and menisci at risk

Knee: Function – General

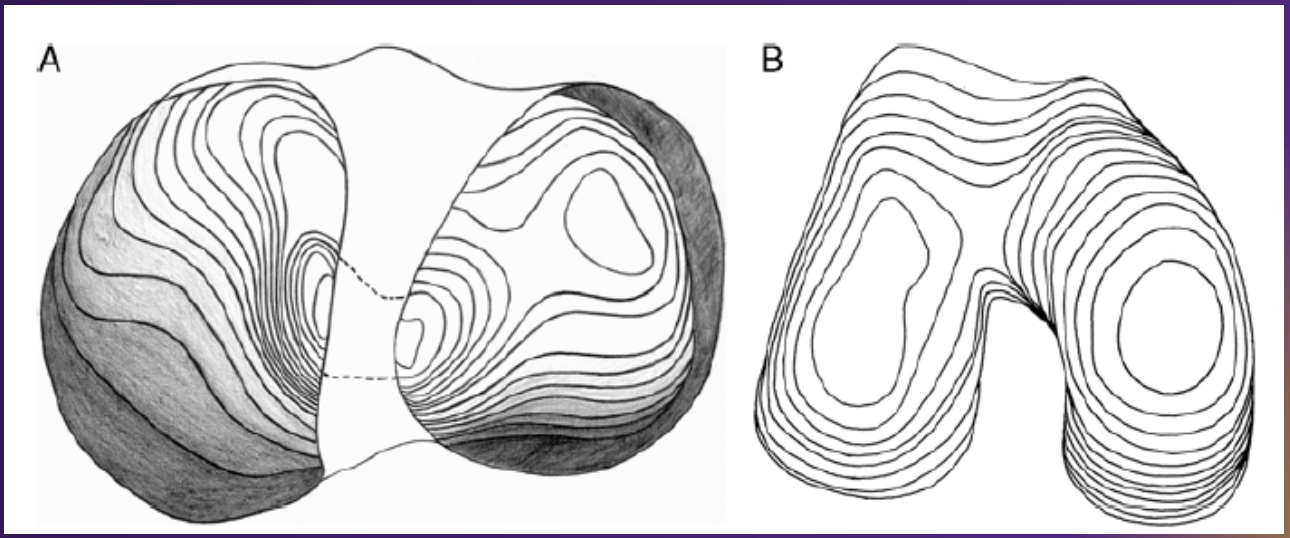
- knee loads during walking (female)
- 3.9x BW level, 8x BW downhill



Kuster, JBJS 1997

Knee: Function – General

- hinge joint – but axis changes, so femur slides on tibia
- screw home mechanism
  - external tibial rotation - knee extends (stance)
    internal tibial rotation - knee flexes (swing)
  - caused by tibial and femoral epicondyles

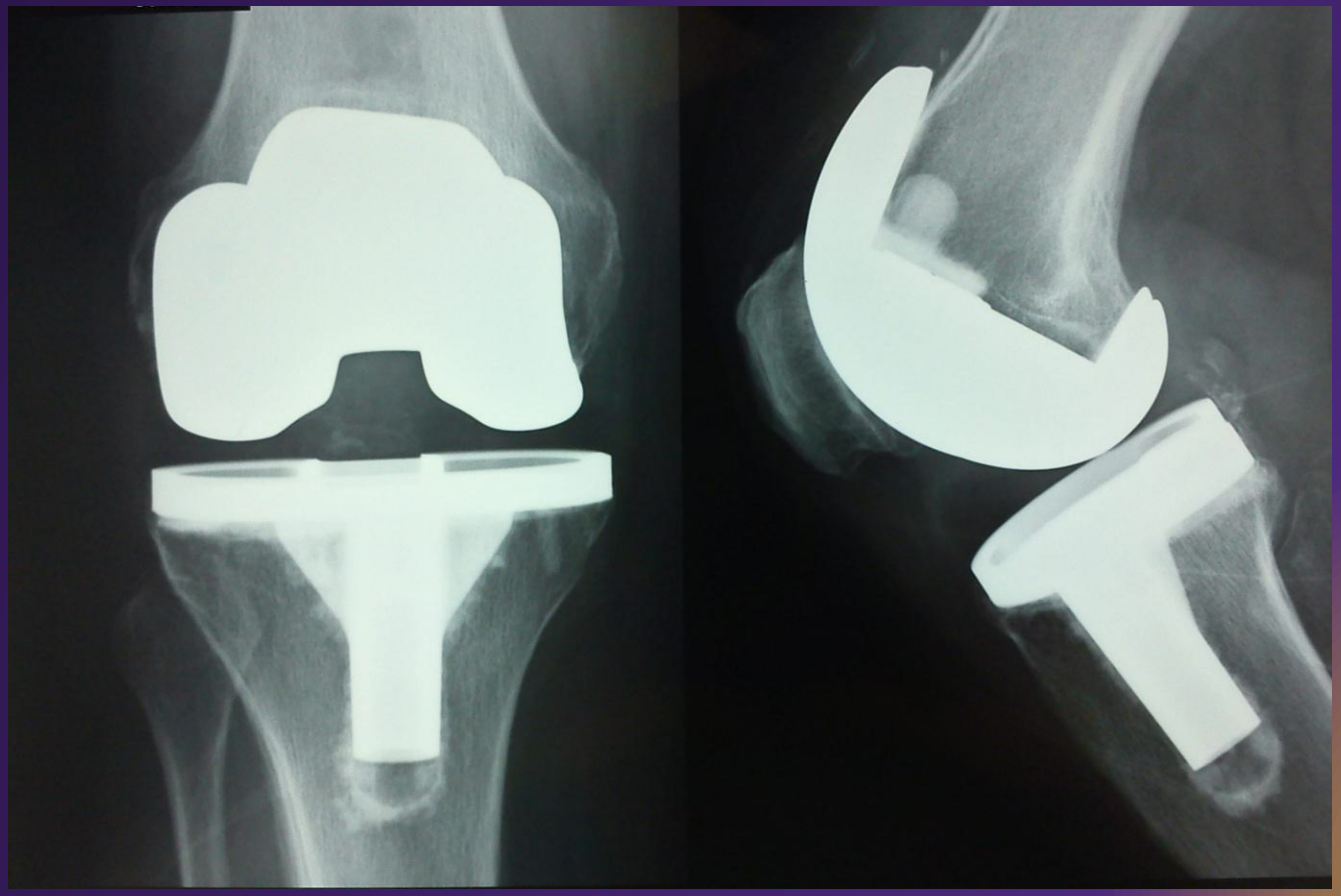


Knee: Pathology - Osteoarthritis

- 4th most common part of body (0.9%), after hands (7.3%), feet (2.3%) and hip (1.5%)
- 94-97% of all TKR due to OA

Knee: Pathology - Arthroplasty

- 719,000 per year in US (National Hospital Discharge Survey: 2010)



Knee: Pathology - Arthroplasty

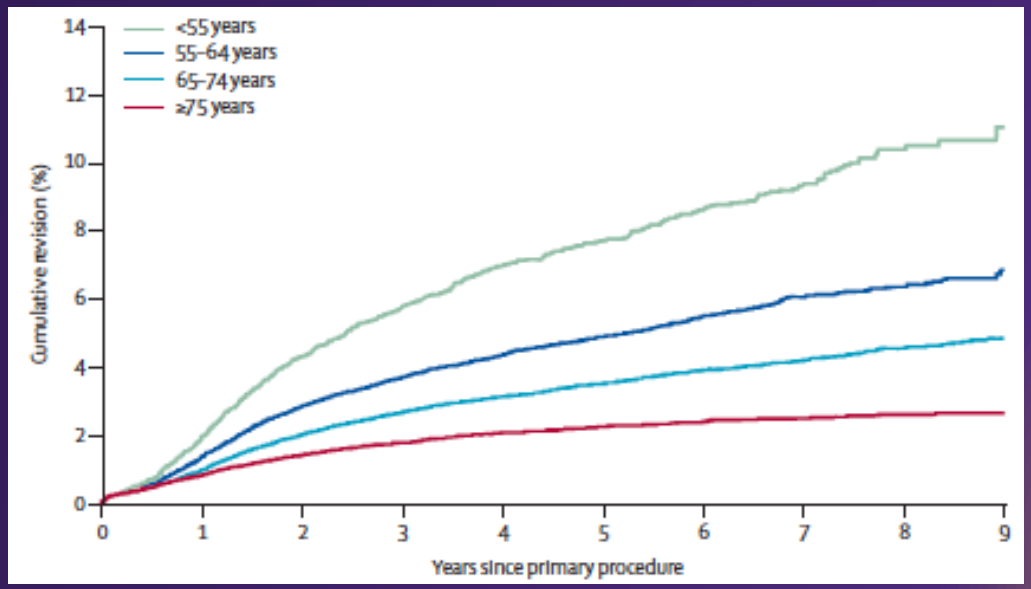
- fixed bearing v. mobile bearing

Knee: Pathology - Arthroplasty

- mobile bearing unicondylar knee replacement

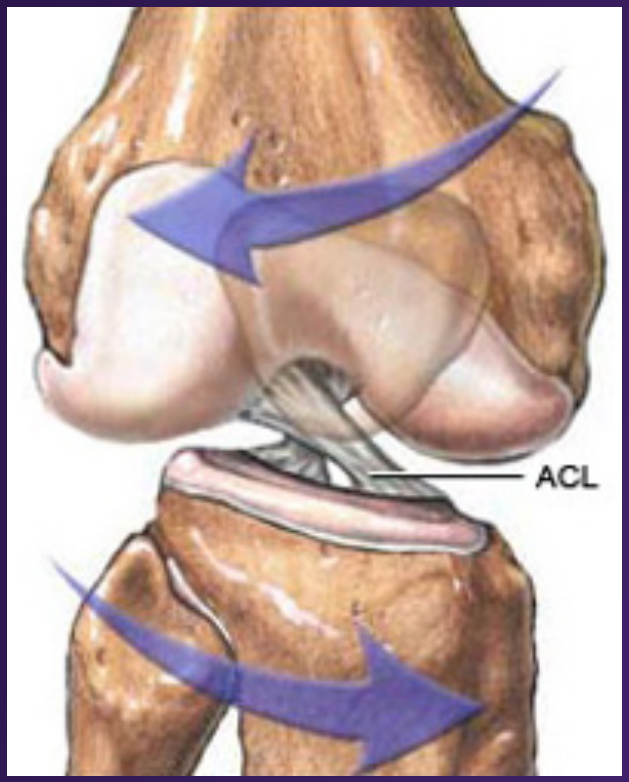
Knee: Pathology - Arthroplasty

- Revision surgery
  - aseptic loosening (30%)
  - infection (20%)
  - pain, instability, stiffness (30-40%)
  - other factors (10-20%)



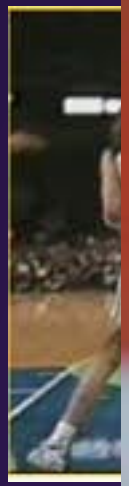
Knee: Pathology – ACL tear

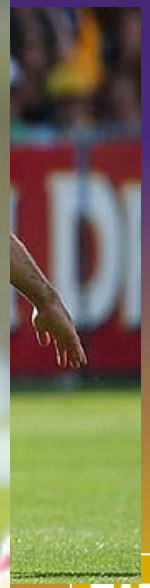
- internal rotation of tibia
- anterior displacement of tibia



Knee: Pathology – ACL tear

-
-



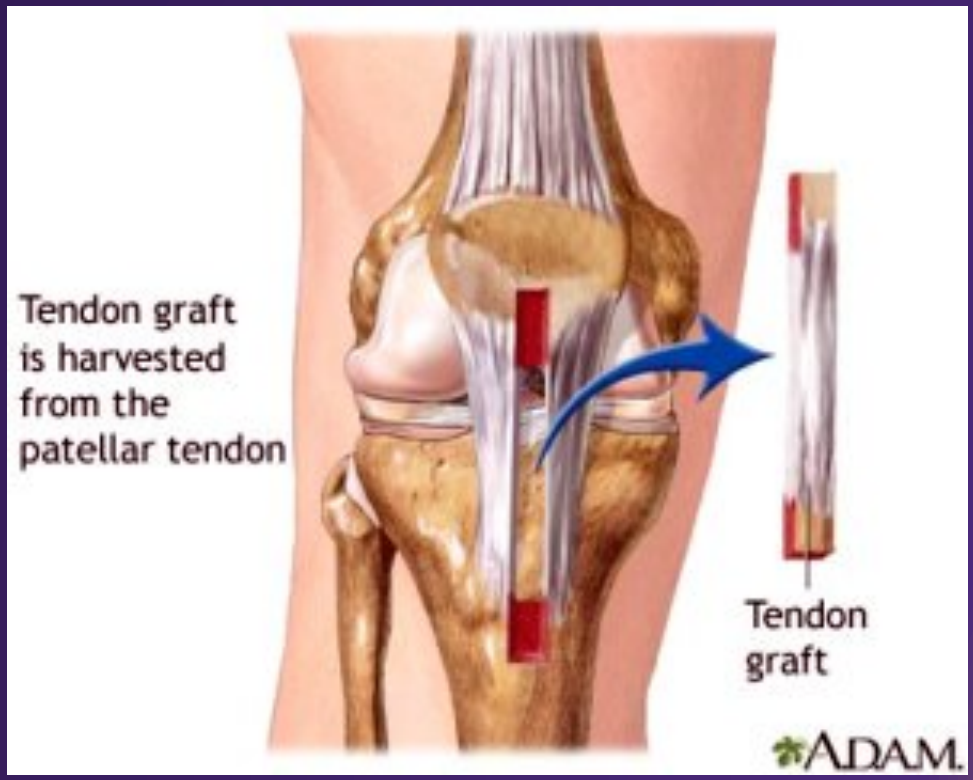


Knee: Pathology – ACL reconstruction

- strengthen knee muscles
  - quads = PCL
  - hamstrings = ACL
- bracing the knee

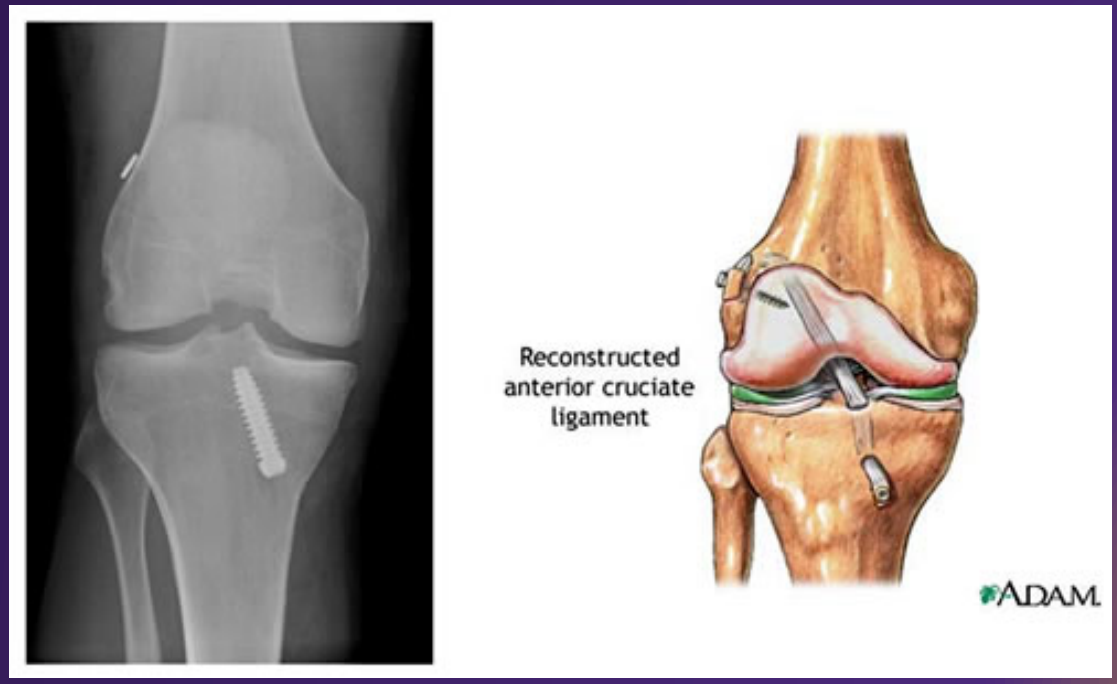
Knee: Pathology – ACL reconstruction

- types of surgery: patellar tendon graft
  - gold standard, 90-95% success
  - donor site pain is main limitation



Knee: Pathology – ACL reconstruction

- types of surgery: hamstring tendon graft
  - interference screw (bioresorbable)
  - longer to heal, harvest is complex, hamstring weakness

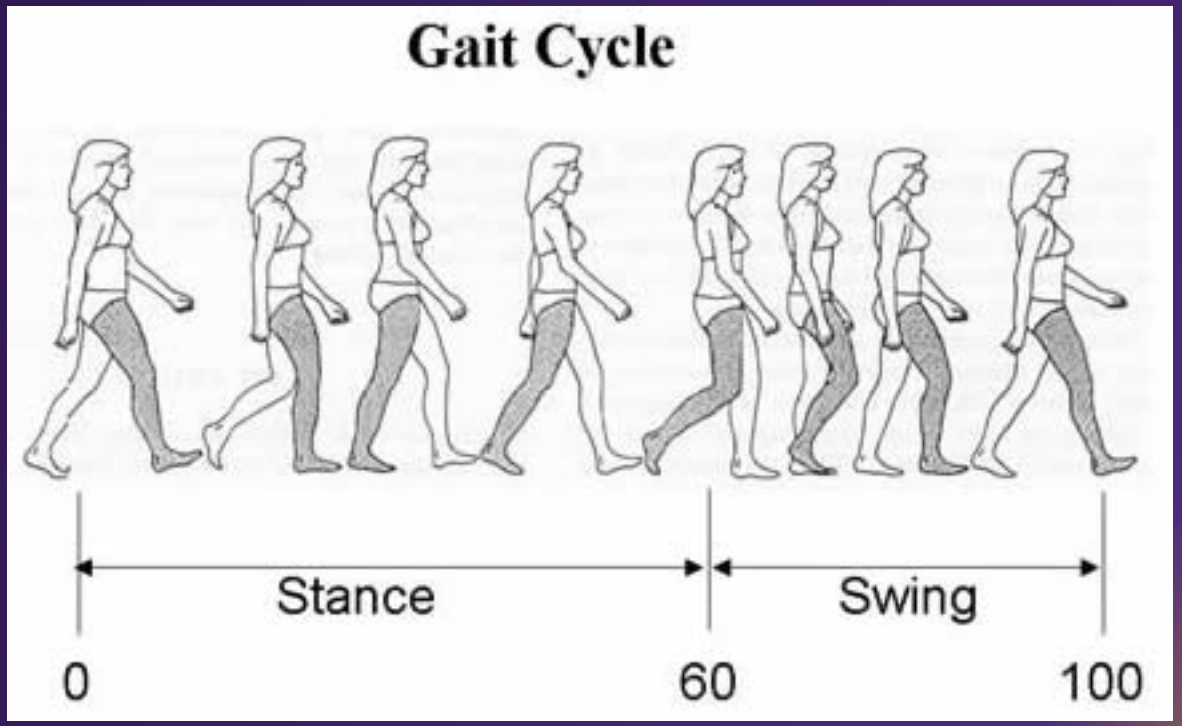


Knee: Pathology – ACL reconstruction

- types of surgery: others?
  - quadriceps tendon graft
    - similar issue to hamstring tendon
  - allograft
    - blood borne disease

Knee Function

- motion data



Knee Function

- motion data